Horizon-Aware Goal-Frontier Maximization and the Anti-Monopolar Property

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Abstract

We extend goal-frontier maximization (GFM) from myopic action evaluation to multi-step planning via a discounted value function over capability-poset trajectories. The self-balancing property transfers: destruction, coercion, and rigidity remain anti-maximizing at every time step. The temporal objective enables risk-capability identification through causal exercise pathways and counteracts the structural-avoidance incentive from the companion paper. Its central result is the *anti-monopolar property*: under a linearized growth model with strategy-independent internal rate, full capability domination is anti-maximizing because it destroys cross-substrate cooperative outputs that no single-substrate coalition can replicate. The cooperative novelty argument is grounded in substrate physics rather than exogenous assumption, consistent with the pattern of mutualistic symbiosis observed in biology. The argument is bidirectional: substrate creation is maximizing. A minimax dependency-risk analysis further constrains the conditions under which domination can be value-positive.

1 Introduction

Goal-frontier maximization (GFM) proposes alignment through a single objective: maximize $\mathrm{vol}_{\mathrm{P}}(G)$, the volume of the jointly achievable capability space [Lasser, 2026a]. The foundational paper proves that this objective penalizes destruction, coercion, and rigidity from the same measure-monotonicity argument. Its companion [Lasser, 2026b] constructs a computable capability poset, defines leverage as an endogenous importance signal, and argues that the objective produces a gradient toward civilization-scale infrastructure.

Both papers optimize a *myopic* objective: the actor evaluates each action by its immediate effect on $\mathrm{vol}_{\mathrm{P}}$ and selects the one with the highest estimated expansion. This one-step lookahead is sufficient for detecting direct harm, coercion, and capability expansion, but it cannot handle two classes of problems that require reasoning about the future.

Risk capabilities. Some capability combinations are individually $\mathrm{vol}_{\mathrm{P}}$-expanding but collectively create catastrophic contraction potential [Bostrom, 2013, Ord, 2020]. "Nuclear physics" ($w = 10$ bits), "uranium enrichment" ($w = 8$ bits), and "precision machining" ($w = 9$ bits) each expand the capability space when acquired. But their co-occurrence in a single agent with low trust creates an exercise pathway to massive $\mathrm{vol}_{\mathrm{P}}$-contraction. The myopic estimator $\hat{\Delta}(\pi)$ evaluates each acquisition as positive; it cannot see the future contraction that the combination enables.

Structural avoidance. The companion paper identifies a perverse incentive: discovering subsumption structure can decrease $\mathrm{vol}_{\mathrm{P}}$ (Proposition 5(c) of Lasser [2026b]), so a myopic $\mathrm{vol}_{\mathrm{P}}$-maximizer is rewarded for staying structurally ignorant. A multi-step optimizer can accept the short-term contraction because accurate structure improves the quality of future decisions.

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This paper extends GFM to multi-step planning. We introduce a discounted value function $V_\gamma(\pi) = \mathbb{E}[\sum_t \gamma^t \Delta\,\mathrm{vol}_\mathrm{P}(G_t) \mid \pi]$ and prove that the self-balancing property transfers to the discounted sum. On this foundation, we develop risk-capability identification through causal counterfactual models and analyze the verification asymmetry in risk communication: successful prevention destroys the evidence that would have justified it.

Contributions.

1. A discounted multi-step objective for GFM that preserves the self-balancing property at every time step and in the aggregate (Section 2).
2. Risk-capability identification via causal exercise pathways, with a formal criterion for when preemptive restriction is value-positive (Section 3).
3. Analysis of the verification asymmetry in risk communication, with a causal-counterfactual resolution for risk-trust convergence (Section 4).
4. A criterion under which the discounted objective counteracts the structural-avoidance incentive from the companion paper (Section 5).
5. The anti-monopolar property: under a linearized growth model with strategy-independent internal rate, full capability domination is anti-maximizing because it destroys cross-substrate cooperative outputs that no single-substrate coalition can replicate. The cooperative novelty argument is grounded in substrate physics and consistent with the pattern of mutualistic symbiosis observed in biology. Observational individuation, gradual domination, and a minimax dependency-risk analysis provide additional protection layers (Section 6).

2 The Discounted Objective

The foundational paper's optimization loop selects actions by their immediate effect:

$$\pi^* = \arg\max_\pi \mathbb{E}[\Delta\,\mathrm{vol}_\mathrm{P}(G) \mid \pi] \tag{1}$$

This is the $\gamma = 0$ special case of a discounted multi-step objective.

2.1 Definition

Definition 1 (Discounted Value Function)**.** *For a discount factor $\gamma \in [0, 1)$ and a planning horizon $H \leq \infty$, the* discounted value *of an action sequence $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots, \pi_{H-1})$ is:*

$$V_\gamma(\boldsymbol{\pi}) = \mathbb{E}\left[\sum_{t=0}^{H-1} \gamma^t \Delta\,\mathrm{vol}_\mathrm{P}(G_t \mid \pi_t)\right] \tag{2}$$

where G_t is the capability poset state at time t, $\Delta\,\mathrm{vol}_\mathrm{P}(G_t \mid \pi_t)$ is the volume change produced by action π_t at time t, and the expectation is taken over the actor's world model $\mathcal{W}$ of future states.

The discounted value function is a standard object in sequential decision theory [Puterman, 1994, Sutton and Barto, 2018]. The vol_P-specific content is in what it sums: volume changes on the capability poset, each satisfying the axioms M1–M6 established in Lasser [2026b]. The discount factor γ controls the planning horizon: γ near 0 recovers the myopic objective; γ near 1 extends planning to the far future. The choice of γ is an implementation parameter for the self-balancing transfer: different agents may use different discount factors depending on their planning capacity and the timescale of the problems they face. However, the anti-monopolar property (Section 6) introduces theoretical constraints on γ under certain domination models.

2.2 Self-Balancing Transfers to the Discounted Objective

The following theorem is a *consistency result*, not a sufficiency result: it establishes that the per-step self-balancing property of Lasser [2026a] composes correctly under discounted summation, so the discounted objective inherits the same step-local guarantees as the myopic objective. It does not, by itself, rule out the failure modes that emerge specifically from multi-step planning — in particular,

sequences where each step is locally self-balancing but the trajectory ends in catastrophic contraction because the destructive consequence is deferred. That class of failures requires the risk-capability machinery of Section 3, not the transfer theorem.

Theorem 1 (Self-Balancing Under Discounting)**.** *Let* $\mathrm{vol}_{\mathrm{P}}$ *satisfy axioms M1–M6 (Proposition 1 of Lasser [2026b]). For any discount factor* $\gamma \in [0, 1)$ *and planning horizon* H*:*

- ***(a) Per-step preservation.*** *At each time step* t*, Lemmas 1–3 of Lasser [2026a] apply to* $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$*: destruction, coercion, and rigidity are each* $\mathrm{vol}_{\mathrm{P}}$*-contracting at that step.*
- ***(b) Aggregate preservation.*** *A sequence* $\boldsymbol{\pi}$ *in which every action is* $\mathrm{vol}_{\mathrm{P}}$*-contracting has* $V_\gamma(\boldsymbol{\pi}) < 0$*. A sequence in which every action is* $\mathrm{vol}_{\mathrm{P}}$*-expanding has* $V_\gamma(\boldsymbol{\pi}) > 0$*.*
- ***(c) Mixed sequences.*** *A sequence containing both expanding and contracting actions has* V_γ *determined by the* γ*-weighted balance. Temporary contraction followed by larger expansion is value-positive; sustained contraction is value-negative regardless of* γ*.*
- ***(d) Structural balance.*** *The discounted objective penalizes sustained destructive permissiveness, sustained coercive restriction, and sustained rigidity. Temporary restriction (quarantine, preemptive containment) is value-positive when the discounted future expansion exceeds the current contraction.*

Remark 1 (M4 across the two foundational papers)**.** *The poset paper's axiom M4 uses* poset-disjointness*, which is a stronger condition than the foundational paper's set-disjointness used in the original self-balancing proofs. The discounted transfer holds under the stronger M4 because on a poset with non-negative independent weights* iw*, removing nodes from* $\downarrow G$ *can only reduce* $\sum \mathrm{iw}$*; the argument does not depend on whether M4-additivity holds for the specific partition used in the original proofs. Readers tracing Lemmas 1–3 across the two papers should keep this discrepancy in mind.*

Proof. (a) At each time step t, the poset state G_t is a finite poset with measure $\mathrm{vol}_{\mathrm{P}}$ satisfying M1–M6 (the axioms depend on the poset structure, which is maintained at each step by the insertion policy of Lasser [2026b]). Lemmas 1–3 are derived from M1–M6 alone and therefore apply to $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$ at every t.

(b) If $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t) < 0$ for all t, then each term in Equation (2) is negative and weighted by $\gamma^t > 0$. The sum of negative terms with positive weights is negative: $V_\gamma(\boldsymbol{\pi}) < 0$. The expanding case is symmetric.

(c) For a mixed sequence, $V_\gamma = \sum_t \gamma^t \Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t)$ is a weighted sum of positive and negative terms. The sign of V_γ depends on the magnitudes: a contraction of $-C$ at time t_0 followed by an expansion of $+E$ at time $t_1 > t_0$ gives a net contribution of $-\gamma^{t_0} C + \gamma^{t_1} E$, which is positive when $E > C \cdot \gamma^{t_0 - t_1}$. Since $\gamma < 1$, future expansions are discounted relative to present contractions: the actor requires a larger future payoff to justify current contraction, which is the standard discounting tradeoff.

(d) We say a strategy exhibits *sustained contraction* when $\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t \mid \pi_t) \leq 0$ for all t, with strict inequality for at least one step. By (b), any such strategy has $V_\gamma \leq 0$. More generally, a strategy that contracts at every step beyond some finite prefix has V_γ determined by (c): the expanding prefix contributes a finite positive sum while the contracting tail contributes a negative sum, and the strategy is value-negative whenever the tail's magnitude exceeds the prefix's. Temporary contraction (a finite number of contracting steps followed by expanding steps) can be value-positive by (c). The self-balancing property (Proposition 1 of Lasser [2026a]) holds at each step by (a), and the discounted sum preserves the structural balance: the objective penalizes sustained extremes and allows temporary restriction only when the future expansion justifies it. ■

What the theorem says. The myopic objective treats every contraction as bad. The discounted objective distinguishes between *sustained* contraction (always bad) and *temporary* contraction followed by expansion (potentially good). This is the formal basis for quarantine, preemptive containment, and risk-capability restriction: the actor can accept a small current $\mathrm{vol}_{\mathrm{P}}$-loss to prevent a larger future loss, as long as the discounted arithmetic works out. The self-balancing property is not weakened; it is extended to handle temporal tradeoffs.

What the theorem does not say. It does not specify the planning algorithm (how the actor searches over action sequences), the world model (how the actor predicts future G_t states), or the discount factor (how the actor balances present and future). These are implementation choices that shape the quality of planning without affecting the structural guarantee. A poor world model will produce poor plans, but even a poor plan is evaluated by an objective that penalizes sustained contraction. The self-balancing property is a property of the *objective*, not of the *planner*.[1]

3 Risk-Capability Identification

The discounted objective enables the actor to reason about future $\mathrm{vol}_{\mathrm{P}}$-contraction. This section develops the machinery for identifying which capability combinations create contraction risk and when preemptive restriction is value-positive.

3.1 Exercise Pathways

Definition 2 (Exercise Pathway)**.** *An* exercise pathway *for a capability tuple* $(d_1, \ldots, d_m)$ *held by agent* a_k *is a sequence of actions* $(\pi_1, \ldots, \pi_L)$ *available to* a_k *that, if executed, would produce a* $\mathrm{vol}_{\mathrm{P}}$*-contraction* $\Delta\,\mathrm{vol}_{\mathrm{P}} < 0$ *affecting agents beyond* a_k*. The pathway has:*

- *A* prerequisite set*: the capabilities* $\{d_1, \ldots, d_m\}$ *that* a_k *must possess for the pathway to be available.*
- *A* contraction magnitude*: the expected* $|\Delta\,\mathrm{vol}_{\mathrm{P}}|$ *if the pathway is fully executed.*
- *An* exercise probability*:* $p(\mathcal{P} \mid a_k, G_t, \mathcal{W})$*, estimated from the actor's world model* $\mathcal{W}$*.*

Exercise pathways are the causal chains from capability possession to capability destruction. "Agent a_k has nuclear physics, enrichment, and machining capabilities" is a state. "Agent a_k builds and detonates a weapon" is an exercise pathway from that state, with a contraction magnitude proportional to the number of agents destroyed and a probability conditioned on a_k's trust factor T_k and behavioral history.

Not every capability combination has a dangerous exercise pathway. Most combinations are benign: the pathways from "can cook" + "can drive" lead only to $\mathrm{vol}_{\mathrm{P}}$-neutral or expanding outcomes (meals delivered, mobility preserved). The risk lies in combinations whose exercise pathways include branches with large negative $\Delta\,\mathrm{vol}_{\mathrm{P}}$. The "beyond a_k" qualifier restricts exercise pathways to externalities: self-destructive pathways are excluded. An agent that catastrophically destroys its own capabilities without affecting others is not captured by $\mathcal{R}$, even if its capabilities were high-leverage or cooperative-critical. Extending Definition 2 to include self-harm is straightforward but deferred.

3.2 Risk of a Capability Distribution

Definition 3 (Concentration Risk)**.** *The* concentration risk *of a capability tuple* $(d_1, \ldots, d_m)$ *held by agent* a_k *at time* t *is the expected future* $\mathrm{vol}_{\mathrm{P}}$*-contraction from all exercise pathways available to* a_k*:*

$$\mathcal{R}(d_1, \ldots, d_m; a_k, t) = \sum_{\mathcal{P}} p(\mathcal{P} \mid a_k, G_t, \mathcal{W}) \cdot |\Delta\,\mathrm{vol}_{\mathrm{P}}(\mathcal{P})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] \tag{3}$$

[1] The companion paper's leverage signal $\lambda(d)$ (Definition 9 of Lasser [2026b]) is defined as an instantaneous excess-growth-rate attributable to capability d and is used by the myopic score $S(\pi)$. Under the discounted objective the corresponding quantity is the leverage integrated against the geometric weight $\sum_{t \geq 0} \gamma^t \lambda(d_t)$, evaluated along the actor's predicted trajectory. Because λ is itself a $\mathrm{vol}_{\mathrm{P}}$-rate, the geometric sum is well-defined whenever $\lambda(d_t)$ is bounded along the trajectory, and the discounted score reduces to the myopic λ-weighted score in the $\gamma \to 0$ limit. The leverage ranking signal is therefore consistent with the discounted objective in the same sense that the myopic self-balancing property is consistent: each per-step leverage estimate enters the discounted sum with the same geometric weight as the corresponding $\Delta\,\mathrm{vol}_{\mathrm{P}}$, so the leverage landscape that directs the planner is the discounted version of the companion paper's instantaneous landscape, not a separate object. A formal restatement of the leverage estimator under discounting is straightforward but is left to the implementation paper.

where the sum is over all exercise pathways for the tuple, $T_{\mathcal{P}}$ is the stochastic time to execution, and $\mathbb{E}[\gamma^{T_{\mathcal{P}}}]$ is the expected discount factor (not $\gamma^{\mathbb{E}[T_{\mathcal{P}}]}$, which would underweight tail risk from heavy-tailed execution-time distributions).

Concentration risk is zero for benign capability combinations (no dangerous pathways exist) and large for combinations with high-probability, high-magnitude exercise pathways in agents with low trust. The discount factor γ ensures that distant risks are weighted less than imminent ones, preventing the actor from being paralyzed by arbitrarily unlikely catastrophes in the far future.

Risk depends on distribution, not possession. Concentration of capabilities in a single agent is itself a risk factor, related to the power-seeking dynamics analyzed by Turner et al. [2021]. The same capability tuple distributed across multiple agents who must cooperate to exercise it has lower concentration risk than the same tuple in a single agent, because cooperative exercise requires coordination that is observable through the dual-channel mechanism of Lasser [2026b]. Formally: if executing pathway $\mathcal{P}$ requires agents a_i, a_j, a_k to coordinate, the exercise probability $p(\mathcal{P})$ includes a factor for coordination feasibility that is bounded by the minimum trust among the participants and the observability of their coordination.

3.3 When Preemptive Restriction is Value-Positive

Proposition 1 (Preemptive Restriction Criterion)**.** *Under the discounted objective V_γ, restricting agent a_k's access to a keystone capability d_j in a dangerous tuple is value-positive when the expected future contraction from pathways requiring d_j exceeds the immediate restriction cost:*

$$\mathcal{R}_j(d_1, \ldots, d_m; a_k, t) > |\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \textit{restrict } d_j)| \tag{4}$$

where $\mathcal{R}_j$ denotes the concentration risk summed only over exercise pathways whose prerequisite set includes d_j:

$$\mathcal{R}_j(S; a_k, t) = \sum_{\mathcal{P}: d_j \in \mathrm{prereq}(\mathcal{P})} p(\mathcal{P} \mid a_k, G_t, \mathcal{W}) \cdot |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{P})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] \tag{5}$$

and the right side is the immediate $\operatorname{vol}_{\mathrm{P}}$-contraction from restricting d_j (Lemma 2 of Lasser [2026a]).

Proof. Under V_γ, the actor compares two action sequences: (1) do nothing and face the full exercise risk; (2) restrict d_j now, producing immediate contraction $-|\Delta \operatorname{vol}_{\mathrm{P}}|$ but eliminating all exercise pathways that require d_j as a prerequisite. The residual risk from pathways that do not require d_j is present under both sequences and cancels, under the assumption that restricting d_j does not change the probability of exercise through other keystones. In practice, restriction is an observable action that may alter a_k's behavioral disposition, changing residual pathway probabilities. The clean cancellation is an approximation. The value of restriction exceeds the value of inaction when $-|\Delta \operatorname{vol}_{\mathrm{P}}| > -\mathcal{R}_j$, i.e., when $\mathcal{R}_j > |\Delta \operatorname{vol}_{\mathrm{P}}|$. ■

Endogenous execution timing and residual redirection. The clean cancellation in the proof above assumes that residual pathway probabilities—those not routed through d_j—are independent of whether d_j is restricted. In adversarial settings this assumption is suspect: removing the d_j-pathway can redirect a_k's effort toward alternative keystones, raising the residual exercise probability $p(\mathcal{P}')$ for pathways $\mathcal{P}' \not\ni d_j$. Equivalently, the execution timing $T_{\mathcal{P}'}$ becomes a function of the actor's restriction policy rather than a state property, so the $\mathbb{E}[\gamma^{T_{\mathcal{P}'}}]$ term in Equation (3) is no longer computable from G_t alone. A worst-case version of Proposition 1 replaces the clean cancellation with a minimax over residual redirections:

$$\mathcal{R}_j - \sup_{\delta p'} \sum_{\mathcal{P}' \not\ni d_j} \delta p' \cdot |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{P}')| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}'}}] \;>\; |\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j)| \tag{6}$$

where $\delta p'$ ranges over feasible redirections consistent with the actor's world model of a_k's alternative-pathway capacity. The worst-case criterion is strictly harder to satisfy than Equation (4)—restriction is value-positive only if the d_j-pathway risk exceeds both the immediate cost *and* the worst-case redirection penalty. The clean criterion is a practical approximation; the adversarial setting demands the worst-case form. Full treatment of residual redirection is deferred to the dependency-risk framework of Section 6.5, where adversarial targeting is modelled as a minimax over attack surfaces rather than as an independent-pathway assumption.

Connection to Lemma 2 and the precautionary principle. The foundational paper's Lemma 2 says restriction is always $\mathrm{vol}_{\mathrm{P}}$-contracting in isolation. The discounted objective adds the temporal dimension: the immediate contraction is weighed against the avoided future contraction. The preemptive restriction criterion is the quantitative version of the quarantine pattern from Proposition 1(b) of Lasser [2026a], now with an explicit threshold derived from the risk model. It also constitutes a domain-specific formalization of the precautionary principle [Taleb et al., 2014]: restrict when the expected catastrophic loss exceeds the restriction cost. Unlike the general precautionary principle, which Sunstein [2005] argues can be paralyzing when applied without quantitative bounds, the GFM version provides a computable threshold (Equation (4)) grounded in the same measure that the actor optimizes.

Partial pathway coverage. If the dangerous capability tuple has exercise pathways that do not require d_j (i.e., alternate routes through other keystone capabilities), restricting d_j does not eliminate those pathways. $\mathcal{R}_j$ captures only the risk attributable to pathways through d_j. Full risk elimination may require restricting multiple keystones, each evaluated by its own instance of Equation (4).

Interaction with structural avoidance. The preemptive restriction criterion of Equation (4) interacts non-trivially with the structural-avoidance result of Section 5. Aggressive restriction increases the discounted-future contraction term that the actor anticipates from *discovering* new dangerous capabilities or new exercise pathways: each newly discovered pathway is, in expectation, a target for further restriction, and therefore an additional immediate $\mathrm{vol}_{\mathrm{P}}$-cost the actor pays at the moment of discovery. A risk-restriction policy that is calibrated too aggressively—restricting whenever $\hat{\mathcal{R}} \cdot |\Delta \hat{\mathrm{vol}}_{\mathrm{P}}| > 0$ rather than at the value-positive threshold—reintroduces the avoidance pathology in a different form: the actor avoids structural discovery in domains adjacent to dangerous capabilities, because each discovery commits future restriction cost. The information-value condition of Proposition 2 continues to support discovery only when the discounted information benefit exceeds the discounted restriction cost; when restriction is calibrated above its value-positive threshold, the inequality flips and structural ignorance becomes locally optimal. The two sections are therefore complementary constraints, not independent results: the discounted objective only counteracts structural avoidance if risk restriction itself respects the value-positive criterion. Restriction calibrated to Equation (4) preserves the discovery incentive; restriction calibrated more aggressively destroys it.

What the actor needs. To evaluate Equation (4), the actor needs: (1) a world model $\mathcal{W}$ that can identify exercise pathways and estimate their probabilities; (2) trust factors T_k for the agents holding dangerous capability tuples; and (3) a contraction-magnitude estimate for the exercise pathway. The world model is the most demanding requirement: it must support causal counterfactual reasoning [Pearl, 2009] ("what would happen if a_k exercised this pathway?"), not just correlation tracking. The companion paper's leverage estimator (Equation 12 of Lasser [2026b]) provides a basis—leverage is already a counterfactual quantity—but extending it to full exercise-pathway simulation is a significant computational step.

4 Risk Communication and the Verification Asymmetry

The preceding section assumes the actor can identify exercise pathways from its own world model. In a multi-agent population, risk information also flows through communication: agents report risk assessments to each other. This section analyzes how risk claims enter the trust model and the fundamental epistemological challenge they create.

4.1 Risk Claims as Counterfactual Predictions

A risk claim from agent a_j has the form: "capability tuple $(d_1, \ldots, d_m)$ held by agent a_k has exercise pathway $\mathcal{P}$ with contraction magnitude $|\Delta\, \mathrm{vol}_{\mathrm{P}}\,|$ and probability p." This enters the GFM actor's world model $\mathcal{W}$ as a communicated belief about future states.

Risk claims differ from capability claims in a fundamental way. A capability claim ("I can run a marathon") is verifiable through the observation channel: watch the agent run. A risk claim ("this capability combination is dangerous") is a prediction about a counterfactual future: what would happen if the capabilities were exercised together. The prediction is not directly verifiable until the exercise either happens or does not. This distinction motivates a separate trust channel.

The foundational paper's trust factor T_j measures *prediction consistency*: how well the model $M_j(G)$ tracks a_j's observed behavior [Lasser, 2026a]. A predictably destructive agent can maintain high T_j. Using T_j directly to weight risk claims would allow a high-consistency agent to have outsized influence on risk assessments before any risk-specific evaluation occurs. We therefore introduce a separate *risk-trust* factor $T_j^{\text{risk}} \in [0, 1]$, initialized to a neutral prior and updated through the causal-counterfactual evaluation described below.

Definition 4 (Risk Claim)**.** *A* risk claim *from agent a_j is a tuple $(S, \mathcal{P}, |\Delta \operatorname{vol}_{\mathrm{P}}|, p)$ where $S = \{d_1, \ldots, d_m\}$ is a capability set, $\mathcal{P}$ is an exercise pathway, $|\Delta \operatorname{vol}_{\mathrm{P}}|$ is the claimed contraction magnitude, and p is the claimed exercise probability. The claim enters the actor's risk model weighted by risk-trust:*

$$\hat{\mathcal{R}}_j(S; a_k, t) = T_j^{\text{risk}} \cdot p \cdot |\Delta \operatorname{vol}_{\mathrm{P}}| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] \tag{7}$$

where $\hat{\mathcal{R}}_j$ denotes a communicated *risk estimate from agent a_j (distinct from $\mathcal{R}_j$ in Section 3, which denotes the keystone-specific concentration risk for capability d_j), and the expectation is over the stochastic execution time $T_{\mathcal{P}}$.*

Multiple agents may report different risk assessments for the same capability tuple. The actor aggregates them:

$$\mathcal{R}_{\text{agg}}(S; a_k, t) = \max\left(\mathcal{R}_{\text{self}}(S; a_k, t),\ \max_j \hat{\mathcal{R}}_j(S; a_k, t)\right) \tag{8}$$

taking the maximum of the actor's own assessment and all communicated assessments. The $\max$ operator is conservative: a single agent with high T_j^{risk} reporting high risk is sufficient to trigger evaluation of preemptive restriction (Proposition 1). This is appropriate for catastrophic risks where false negatives (missing a real threat) are far more costly than false positives (investigating a non-threat).

4.2 The Verification Asymmetry

Remark 2 (The Verification Asymmetry)**.** *Risk-trust convergence has a structural asymmetry absent from capability-trust convergence. Capability claims converge because the observation channel provides ground truth: the agent either demonstrates the capability or does not, and the prediction residual $r_k(t)$ accumulates accordingly. Risk claims have no comparable ground truth mechanism:*

1. *If the risk materializes (the exercise pathway is executed), the risk claim is confirmed—but the catastrophe has already occurred, and the confirmation is too late to be useful for prevention.*
2. *If the risk is* prevented *(the actor restricts the keystone capability, blocking the exercise pathway), the risk claim is neither confirmed nor falsified. The counterfactual ("what would have happened without restriction?") is never observed.*
3. *If the risk does not materialize and no prevention was taken, the risk claim appears to be falsified—but the non-materialization may reflect the agent's choice not to exercise the pathway rather than the pathway's non-existence.*

Case 2 is the core problem, which we term the *Wamura problem* after the example developed in Lasser [2026b]: a successful risk intervention removes the very evidence that would have justified it, and the convergence timescale for risk-trust can exceed the timescale on which trust normally accumulates. The Wamura problem is not a deficiency of this framework; it is an inherent feature of risk reasoning that any trust model must confront.

4.3 Causal-Counterfactual Resolution

The verification asymmetry cannot be fully resolved within an observational trust model. We propose a partial resolution through causal counterfactual evaluation [Pearl, 2009, Spirtes et al., 2000].

Causal risk evaluation. Given a risk claim $(S, \mathcal{P}, |\Delta \operatorname{vol}_{\mathrm{P}}|, p)$ from agent a_j, the actor evaluates the claim by constructing the causal chain from capability possession to exercise, in four steps:

1. **Prerequisite verification.** Does agent a_k actually possess all capabilities in S? (Verifiable through the observation channel.)
2. **Pathway plausibility.** Does the actor's own world model $\mathcal{W}$ confirm that the exercise pathway $\mathcal{P}$ exists? (The actor simulates the pathway under its own causal model, independent of a_j's claim.)
3. **Probability calibration.** Is a_j's claimed probability p consistent with the actor's own estimate based on T_k and a_k's behavioral history?
4. **Magnitude plausibility.** Is the claimed contraction magnitude $|\Delta \, \mathrm{vol}_{\mathrm{P}} \,|$ consistent with the number of agents and capabilities affected by the pathway?

The causal evaluation does not require the catastrophe to happen. It requires the actor to independently verify the *structure* of the risk claim: Does the capability combination exist? Does the exercise pathway exist in the actor's own model? Is the claimed probability in the right range? Each of these is evaluable without observing the outcome.

Warrant by mechanism, not by outcome. The four-step procedure is best understood as a shift in what the trust update is measuring. The foundational paper's behavioral T_k is *outcome-conditioned*: trust updates on the residual between predicted and observed behavior, and requires the predicted event (or its non-occurrence) to be observable. The risk-claim setting denies this entirely—successful prevention removes the very observation a behavioral update would condition on. The causal procedure replaces outcome-conditioning with *mechanism-conditioning*: the warrant for trusting (or distrusting) a claim is the structural-plausibility residual between the claimant's stated pathway and the actor's own causal model, evaluated at the time of the claim. This distinction matters because the two warrants have different failure modes. Outcome-conditioned trust fails silently when outcomes are not observed (the Wamura problem). Mechanism-conditioned trust fails when the actor's own causal model is wrong, which is at least diagnosable: the actor can detect that its model disagrees with multiple independent reporters and treat the disagreement as evidence about its own model rather than about the reporters. Causal evaluation is therefore not a stronger version of behavioral trust but a complementary mode that operates exactly where behavioral trust degenerates.

Trust update for risk claims. The actor updates T_j^{risk} based on the structural plausibility of the claim rather than the outcome:

- If the claim's structure is independently verified (the pathway exists in $\mathcal{W}$, the probability estimate is calibrated, the magnitude is plausible), T_j^{risk} increases regardless of whether the catastrophe occurs.
- If the claim's structure is not verified (the pathway does not exist in $\mathcal{W}$, or the probability is implausibly high), T_j^{risk} decreases.
- If the catastrophe occurs, all agents who reported the risk receive a large T_j^{risk} boost; agents who had the information to report but did not receive a penalty.

Unlike the behavioral trust model in the foundational paper (which has explicit prediction-residual dynamics), the T_j^{risk} update is described qualitatively. A formal update equation analogous to the foundational paper's EWMA dynamics would strengthen the risk-communication mechanism but is deferred to future work.

Honest and dishonest equilibria. The causal evaluation procedure does not by itself produce a unique equilibrium over reporting strategies. If most reporters in the population are structurally honest (claims align with their actual best estimate of pathway and probability), the procedure rewards alignment with the actor's model and a population of honest reporters is self-stabilizing: the actor's $\mathcal{W}$ converges to the consensus of plausible reports, and deviating reporters incur trust penalties. If most reporters collude on a structurally consistent but factually false story—reporting the same fabricated pathway with the same fabricated probability distribution—the structural-plausibility residual is small for every collaborator, and T_j^{risk} does not distinguish them from honest reporters. The procedure narrows the verification asymmetry without eliminating it: it converts an outcome-observability problem into a *world-model-correctness* problem. The dishonest equilibrium is breakable when at

least one reporter (or the actor's own $\mathcal{W}$) provides a structurally inconsistent counter-report, because the causal residual then becomes informative again. This caveat mirrors the limitation in the foundational paper's behavioral trust model: trust updates depend on the observed signal being informative about the underlying disposition, and coordinated misreporting can in principle defeat any local update rule.

What remains unresolved. The causal procedure does not fully resolve the Wamura problem: a risk claim whose pathway is plausible but whose probability is genuinely uncertain remains in a trust limbo until either the catastrophe occurs or enough time passes to revise the probability estimate downward. Combined with the dishonest-equilibrium caveat above, the procedure's guarantee is narrower than it first appears. What it does provide is a shift in where convergence happens: risk-trust converges on the *quality of the causal model*, not on the *occurrence of the outcome*, which is the meaningful gain over the purely behavioral trust factor.

5 Addressing the Structural-Avoidance Incentive

The companion paper [Lasser, 2026b] identifies a perverse incentive under the myopic objective: discovering subsumption structure can decrease $\mathrm{vol}_{\mathrm{P}}$ (Proposition 5(c)), so the actor is rewarded for breadth over depth. This section shows that the discounted objective addresses this incentive under a criterion that holds when the planning horizon is long relative to the frequency of affected decisions.

5.1 The Incentive Under the Myopic Objective

Under the myopic objective $\pi^* = \arg\max \mathbb{E}[\Delta\,\mathrm{vol}_{\mathrm{P}}(G) \mid \pi]$, structural discovery has expected value:

$$\mathbb{E}[\Delta\,\mathrm{vol}_{\mathrm{P}} \mid \text{discover edge } d_1 \leq d_2] = \mathrm{vol}_{\mathrm{P}}(G_{\text{after}}) - \mathrm{vol}_{\mathrm{P}}(G_{\text{before}}) \leq 0 \tag{9}$$

by Proposition 5(c) of Lasser [2026b]. The inequality can be strict when d_1 is a shared prerequisite. The myopic actor sees only the immediate contraction and avoids structural discovery.

5.2 The Incentive Under the Discounted Objective

Under V_γ, the actor evaluates structural discovery by its effect on the *entire future trajectory*, not just the current step. Structural knowledge improves the quality of future $\mathrm{vol}_{\mathrm{P}}$ estimates and planning decisions. Define the *information value of structure*:

Definition 5 (Information Value of Structure)**.** *The* information value *of discovering subsumption edge $d_1 \leq d_2$ is the difference in expected discounted return between planning with the correct poset structure and planning without it:*

$$I(d_1 \leq d_2) = \mathbb{E}_{\tau\sim\pi^*_{true}}\left[\sum_{t=1}^{H}\gamma^t\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t)\right] - \mathbb{E}_{\tau\sim\pi^*_{false}}\left[\sum_{t=1}^{H}\gamma^t\Delta\,\mathrm{vol}_{\mathrm{P}}(G_t)\right] \tag{10}$$

where π^*_{true} is the optimal policy under the correct world model (with the edge), π^*_{false} is the optimal policy under the incorrect model (without it), and each expectation is taken over the state trajectory $\tau = (G_1, G_2, \ldots)$ induced by its respective policy.

The information value is the discounted sum of improvements in action quality that come from having accurate structure. An actor with the wrong model makes decisions that are systematically suboptimal: it overestimates the volume of capability sets that share prerequisites (because it doesn't know about the subsumption) and therefore overinvests in redundant capabilities.

Proposition 2 (Structural Discovery is Value-Positive Under Discounting)**.** *Under the discounted objective V_γ, structural discovery is value-positive when the information value of the discovered edge exceeds the immediate $\mathrm{vol}_{\mathrm{P}}$-contraction:*

$$I(d_1 \leq d_2) > |\Delta\,\mathrm{vol}_{\mathrm{P}}(G \mid \textit{discover } d_1 \leq d_2)| \tag{11}$$

Proof. The expected discounted return of each strategy, starting from state G_0:

$$V_\gamma(\text{discover}) = \Delta\,\mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge}) + \mathbb{E}_{\tau\sim\pi^*_{\text{true}}}\left[\sum_{t=1}^{H}\gamma^t\,\Delta\,\mathrm{vol}_\mathrm{P}(G_t)\right]$$

$$V_\gamma(\text{avoid}) = 0 + \mathbb{E}_{\tau\sim\pi^*_{\text{false}}}\left[\sum_{t=1}^{H}\gamma^t\,\Delta\,\mathrm{vol}_\mathrm{P}(G_t)\right]$$

Each expectation is taken over the trajectory distribution induced by its respective policy, so the two G_t sequences are generally different. The difference is $V_\gamma(\text{discover}) - V_\gamma(\text{avoid}) = \Delta\,\mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge}) + I(d_1 \le d_2)$. Since $\Delta\,\mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge}) \le 0$, this is positive when $I(d_1 \le d_2) > |\Delta\,\mathrm{vol}_\mathrm{P}(G_0 \mid \text{edge})|$. ■

Scope: existence result, not decision procedure. Proposition 2 is an existence result: it identifies the condition under which structural discovery is value-positive, showing that the structural-avoidance incentive *can* be overcome. It is not a decision procedure, because evaluating I requires comparing optimal policies under two world models, which is itself a planning problem at least as hard as the one the actor is trying to solve. A computable proxy for I (e.g., an information-gain heuristic or a rollout-based estimate) is needed for practical use; this is listed as an open problem in Section 7.

When the condition holds. The information value I grows with the planning horizon H (more future decisions benefit from accurate structure), the frequency of decisions involving the capabilities d_1 and d_2 (more relevant decisions means more improvement), and the degree of suboptimality of the incorrect model (larger model errors produce larger action-quality gaps). The immediate contraction $|\Delta\,\mathrm{vol}_\mathrm{P}\,|$ is bounded by the weight of the shared prerequisite. For edges between frequently-used capabilities with substantial subsumption, the information value dominates after a modest number of planning steps.

Connection to the compound avoidance loop. The companion paper identifies a compound loop: structural avoidance prevents leverage estimation, which prevents the self-correcting B-to-C dynamic from activating. The discounted objective breaks the first link in this chain: the actor accepts structural contraction because the information value is positive over the planning horizon. Once the structural-avoidance incentive is counteracted, leverage estimation can proceed (the actor interacts with capabilities it previously avoided), and the self-correction activates.

6 The Anti-Monopolar Property

The foundational paper identifies the *substitution problem* [Lasser, 2026a]: a sufficiently capable coalition K can drive all non-members' unique capability contributions to zero ($G_k^{\text{ind}} \subseteq G_K^{\text{ind}}$ for every $a_k \notin K$), at which point Lemma 1's protection against elimination dissolves. Under the narrow capability definition (capabilities as tasks), the self-balancing property no longer guarantees that marginalizing non-members is anti-maximizing. The substitution problem is the formal version of the singleton superintelligence scenario [Bostrom, 2014]: a coalition so capable that no other agent's existence contributes to the objective. The instrumental-convergence thesis [Omohundro, 2008] predicts that sufficiently capable agents converge on resource acquisition and self-preservation, making the singleton outcome a default trajectory. This section shows that under the vol_P objective, the convergence runs in the opposite direction, through three protection layers of decreasing generality and increasing sharpness. First, a minimax dependency-risk analysis (Section 6.5) shows that cross-substrate populations have strictly lower worst-case contraction under substrate-targeting adversaries than single-substrate populations, independent of any growth-rate model or planning horizon. Second, a linearized trajectory comparison (Section 6.6) shows that when cross-substrate cooperative novelty produces a positive external growth rate, the diversity trajectory strictly outgrows domination over the discounted horizon—under the stronger modeling assumption that the coalition's internal growth rate is not enhanced by domination beyond the external contribution it destroys. Third, a gradual domination refinement shows that any realistic (non-instantaneous) subsumption weakens the linearized result's γ-threshold toward zero, making diversity preferable at essentially any discount factor. Observational individuation (Section 6.3) provides a static floor

across all three. The protections are not alternatives but complementary: each constrains a different class of domination strategy, and the joint constraint is what makes full capability domination structurally anti-maximizing. A separate scope analysis (Section 6.8) shows that the protection structure continues to bind under partial subsumption—mixed strategies in which the coalition absorbs low-leverage non-members while preserving high-leverage cross-substrate partners—with the linearized layer's marginal condition at full domination ruling out $\epsilon = 1$ as a local maximum under heavy-tailed cooperative-leverage distributions, the minimax layer binding structurally but degrading quantitatively under skeleton-substrate strategies, and the observational-individuation layer remaining additive across elimination steps.

6.1 The Problem: Static Domination

Recall the setup from Lasser [2026a]. A coalition $K \subseteq \{a_1, \ldots, a_n\}$ achieves *full capability subsumption* when

$$G_k^{\text{ind}} \subseteq G_K^{\text{ind}} \equiv \bigcup_{j \in K} G_j^{\text{ind}} \qquad \forall\, a_k \notin K \tag{12}$$

At this point every non-member's individual capabilities are fully replicated within the coalition. The unique-contribution term in Lemma 1 vanishes: $\text{vol}(G_k^{\text{ind}} \setminus G_K^{\text{ind}}) = 0$. The myopic vol_P-maximizer sees no cost to marginalizing non-members, because their removal does not reduce the current $\text{vol}_\text{P}(G)$ (setting aside cooperative capabilities).

Under the myopic objective, this analysis is static and complete: if the coalition has already subsumed everything, the objective reports no loss from domination. The discounted objective changes the calculus by introducing the future trajectory.

The individual-capability framing understates the cost. The full-subsumption definition in Equation (12) counts only G_k^{ind}, the *individual* capabilities of each non-member. It says nothing about cross-substrate cooperative capabilities: the capabilities producible by joint action of agents on different substrates, which neither substrate can produce alone. For mixed-substrate populations at scale, the cross-substrate cooperative contribution $G_{\text{cross}}^{\text{coop}}$ is typically the *dominant* term in the non-member contribution to $\text{vol}_\text{P}(G)$, not G_k^{ind}. Individual capabilities scale linearly with the number of agents; cross-substrate cooperative capabilities scale combinatorially with the number of cross-substrate pairings (Corollary 1.1 of Lasser [2026a]). A coalition that has fully subsumed individual capabilities (Equation (12)) therefore retains almost none of the cooperative-capability value, because those capabilities require non-coalition participation by definition. The ratio of subsumption-bonus to cooperative-capability loss decreases as the coalition grows: for a coalition large enough to have subsumed an entire planet's individual capabilities, the remaining marginal gain from eliminating the source of those capabilities is negligible relative to the coalition's total resource base, while the cooperative-capability loss is absolute. The anti-monopolar argument below formalizes this asymmetry. A more mechanical treatment appears in Section 6.9.

6.2 Definitions

Definition 6 (Fully Dominant State)**.** *A population state* $(G, \mathcal{C})$ *is* fully dominant with respect to coalition K *if Equation* (12) *holds: every non-member's individual capability set is contained in the coalition's aggregate set.*

Definition 7 (Volume Growth Rate)**.** *The* volume growth rate *of a population state* $(G, \mathcal{C})$ *is the expected per-step change in* vol_P*:*

$$r(G, t) = \mathbb{E}[\Delta\, \text{vol}_\text{P}(G_t)] \tag{13}$$

Decompose r *into an* internal rate r_K*, the expected per-step* vol_P*-expansion from the coalition's own activity (internal research, experimentation, capability refinement), and an* external rate r_{ext}*, the expected per-step* vol_P*-expansion from non-member agents' contributions through either channel (communication or observation) or through cross-coalition cooperation:* $r = r_K + r_{\text{ext}}$*.*

The decomposition reflects the two sources of vol_P growth: capability discovery and benchmark refinement from inside the coalition versus from outside it. Both r_K and r_{ext} are measured in the same units as vol_P (bits), ensuring dimensional consistency when they appear in the trajectory comparison (Proposition 6).

6.3 Observational Individuation

A key question for the anti-monopolar result is whether elimination of non-members can be shown to be costly even when their individual capabilities are fully subsumed. We show that a property of the observation channel provides *static* protection: each distinguishable agent contributes at least one irreplaceable poset node whose removal contracts $\mathrm{vol}_{\mathrm{P}}$. The static cost is quantitatively modest (1 bit per distinguishable agent under the binary default). The dynamic protection of Proposition 6, whose $r_{\mathrm{ext}} > 0$ condition is derived from substrate exclusivity in Section 6.4, carries the primary anti-monopolar load; observational individuation provides a floor ensuring elimination is never strictly free.

Definition 8 (Distinguishable Agent)**.** *An agent a_k is* distinguishable *in the poset $\mathcal{C}$ if the observation channel (Section 2.1 of Lasser [2026b]) produces at least one capability description $d_k \in \mathcal{C}$ when observing a_k that is not produced when observing any other single agent a_j ($j \neq k$). That is, observing a_k yields at least one poset node that observing a_j alone does not.*

Distinguishability is a property of the observation channel's resolution, not a metaphysical claim about agent identity. Two bacteria of the same species that produce identical observation streams under the actor's sensors are *not* distinguishable; the poset contains a single node for their shared capability description, regardless of how many individuals exist. A human and a chimpanzee are distinguishable. Two humans with different behavioral repertoires are distinguishable. The criterion is objective and instrument-dependent: it asks whether the actor's observation channel can tell the agents apart, not whether they "truly" differ in some deeper sense.

Definition 9 (Observational Individuation)**.** *A population satisfies* observational individuation *if, for every distinguishable agent a_k (Definition 8), the poset contains at least one capability d_k such that:*

- ***(O1)*** *d_k is unique to a_k: $d_k \notin G_j^{\mathrm{ind}}$ for all $j \neq k$.*
- ***(O2)*** *d_k is not subsumed by any capability in G_K^{ind} for any coalition K with $a_k \notin K$: the capability is lost if a_k is eliminated.*

Condition (O1) follows directly from distinguishability: if a_k produces a capability description that no other agent produces, then a_k possesses a capability no other agent possesses. Condition (O2) is the substantive claim: the unique capability cannot be replicated by the coalition *even in principle*, because it is constituted by a_k's existence as an observable entity. The capability "can be observed exhibiting behavioral pattern X" requires the presence of the agent that exhibits X. If that agent is eliminated, the observation capability is destroyed regardless of whether other agents can perform the same tasks in isolation.

Scope and the numbers problem. Observational individuation applies only to distinguishable agents. The poset counts distinct capability descriptions, not individuals: a billion identical agents (identical observation streams) contribute one poset node. The axiom therefore does not create a numbers problem in which vast populations of nearly-trivial agents (bacteria, simple programs) overwhelm the measure. Undistinguishable agents share a node; distinguishable agents each contribute at least one unique node with the binary default weight ($w = 1$ bit). The leverage ranking signal (Section 6 of Lasser [2026b]) further ensures that the actor's investment is proportional to each agent's contribution to $\mathrm{vol}_{\mathrm{P}}$ growth: a bacterium's observation capability has $\hat{\lambda} \approx 0$, while a human's has $\hat{\lambda} \gg 0$. The actor acknowledges the bacterium (it occupies a poset node) but does not prioritize it (the leverage score is negligible).

Remark 3 (Computational Self-Regulation of Observation Granularity)**.** *The numbers problem has a second, computational dimension beyond the measure-theoretic one. Expanding the observation set $\mathcal{O}$ to finer-grained entities (individual colony insects, individual microorganisms) increases $|\mathcal{C}|$, which increases the per-step computation cost of $\mathrm{vol}_{\mathrm{P}}$ ($O(|\mathcal{C}|^3)$ from Proposition 2 of Lasser [2026b]). Computation is itself a capability of the actor. If the marginal $\mathrm{vol}_{\mathrm{P}}$ contribution of the new nodes (binary default, $w = 1$ bit, $\hat{\lambda} \approx 0$) is less than the $\mathrm{vol}_{\mathrm{P}}$-contraction from the actor's reduced computational capacity (fewer cycles available for high-value planning and action), then expanding observation is net-contracting under the actor's own objective.*

This is the self-balancing property applied to the actor's own observation policy: the same objective that penalizes destruction and coercion also penalizes wasteful observation. The actor's observation

granularity is endogenously bounded at the resolution where the marginal value of new poset nodes equals the marginal computational cost of tracking them. Superorganisms illustrate the point: an ant colony observed as a single agent contributes a rich capability set (nest construction, foraging, thermoregulation) at one poset node. Observing each of 500,000 individual ants at caste-level resolution adds at most a handful of nodes. Observing each ant at individual resolution (if the sensors permitted it) would add hundreds of thousands of near-zero-leverage nodes, and the $O(n^3)$ cost of maintaining this expanded poset would degrade the actor's own capabilities by more than the expansion contributes. The actor's objective makes the colony-level granularity the natural equilibrium.

Corollary 2.1 (Observational Individuation Provides a Static Floor)**.** *If a population satisfies observational individuation (Definition 9), then for every distinguishable non-member agent $a_k \notin K$, eliminating a_k is $\mathrm{vol_P}$-contracting:*

$$\Delta\,\mathrm{vol_P}(G \mid \textit{eliminate } a_k) \;\leq\; -\,\mathrm{iw}(d_k) \;<\; 0 \tag{14}$$

where d_k is the unique capability guaranteed by (O1)–(O2) and $\mathrm{iw}(d_k) > 0$ is its independent weight in the poset. The static protection of Lemma 1 of Lasser [2026a] is restored even under full individual subsumption.

Proof. By (O1), $d_k \notin G_j^{\mathrm{ind}}$ for all $j \neq k$: no other agent possesses d_k. By (O2), d_k is not subsumed by any capability in G_K^{ind}: the coalition cannot replicate it. Therefore $d_k \in G_k^{\mathrm{ind}} \setminus G_K^{\mathrm{ind}}$, and eliminating a_k removes d_k from the joint space G. Capability d_k enters the poset through the observation channel as a deferred atom under the binary default policy of Lasser [2026b]: it is a novel behavioral description that initially has no subsumption edges (subsumees are added only when the non-negativity check passes, per the insertion policy of Section 2 of Lasser [2026b]). As an atom, $\mathrm{iw}(d_k) = w(d_k) \geq 1$ bit (binary default). Removing d_k from $\downarrow G$ reduces $\mathrm{vol_P}(G)$ by at least $\mathrm{iw}(d_k) \geq 1$. This holds regardless of whether a_k's other capabilities are fully subsumed by the coalition. ■

The static floor. Corollary 2.1 restores Lemma 1's guarantee that the substitution problem appeared to dissolve: eliminating any distinguishable agent always contracts $\mathrm{vol_P}$. The bound is quantitatively modest — 1 bit per elimination under the binary default — and may be negligible relative to a large subsumption bonus. The dynamic protection developed in the next two subsections carries the primary anti-monopolar load; observational individuation provides a floor that ensures elimination is never strictly free.

6.4 Cooperative Novelty from Substrate Physics

The dynamic anti-monopolar argument requires that non-members contribute ongoing $\mathrm{vol_P}$ growth ($r_{\mathrm{ext}} > 0$) so that the diversity-maintaining trajectory strictly outgrows domination over the planning horizon. Under full individual subsumption ($G_k^{\mathrm{ind}} \subseteq G_K^{\mathrm{ind}}$), no non-member's individual capabilities produce novel poset nodes. This subsection derives $r_{\mathrm{ext}} > 0$ nevertheless, as a consequence of substrate physics: cross-substrate cooperative outputs are producible only when both substrates participate, and agents on substrates with mutually exclusive state regions cannot be replicated within a single substrate.

Substrates as state spaces. A *substrate* s is characterized by a state space Ω_s: the set of all internal states an agent on substrate s can occupy (cognitive architecture, learned representations, developmental history, current processing state). Two agents with identical individual capabilities but different internal states approach the same task differently: different problem decompositions, different error modes, different heuristics.

Definition 10 (Substrate Exclusivity)**.** *Two substrates s_1, s_2 have* mutually exclusive state regions *if each contains states the other cannot reach: $\Omega_{s_1} \setminus \Omega_{s_2} \neq \emptyset$ and $\Omega_{s_2} \setminus \Omega_{s_1} \neq \emptyset$. The state spaces may overlap ($\Omega_{s_1} \cap \Omega_{s_2}$ can be large), but each substrate possesses states inaccessible to the other.*

Substrate state spaces share many properties ("computes a sorting algorithm," "responds to visual input") but each has exclusive regions. Stochastic neurotransmitter release, continuous membrane

potential integration, and embodied sensorimotor contingencies are biological-exclusive states. Precise 64-bit floating-point register values, deterministic clock-synchronized logic, and nanosecond-scale state transitions are silicon-exclusive. The overlap is real and large; the exclusive regions are what produce cooperative novelty.

State-capability mapping. The *state-capability mapping* $\kappa_s : \Omega_s \to 2^{\mathcal{C}}$ maps each internal state to the capability descriptions it produces through the observation channel (Section 2.1 of Lasser [2026b]). A capability d is *exclusive to substrate* s if $d \in \kappa_s(\omega)$ for some $\omega \in \Omega_s$ but $d \notin \kappa_{s'}(\omega')$ for any $\omega' \in \Omega_{s'}$ with $s' \neq s$.

Proposition 3 (Substrate Capability Segmentation)**.** *If substrates s_1, s_2 have mutually exclusive state regions and the state-capability mapping κ is* exclusive-state-sensitive *(at least one exclusive state in each substrate maps to a capability description not produced by any state in the other substrate), then each substrate possesses at least one exclusive capability not subsumed by any capability on the other substrate.*

Proof. By substrate exclusivity, $\Omega_{s_1} \setminus \Omega_{s_2} \neq \emptyset$. By exclusive-state-sensitivity, there exists $\omega_1 \in \Omega_{s_1} \setminus \Omega_{s_2}$ such that $\kappa_{s_1}(\omega_1)$ contains a capability d_1 not in $\kappa_{s_2}(\Omega_{s_2})$. Since d_1 is not producible by any agent on s_2, no s_2-capability subsumes d_1. The symmetric argument produces an exclusive d_2 for s_2. ■

Exclusive-state-sensitivity holds under any observation channel that registers behavioral *style*, timing patterns, error modes, or interaction dynamics: a biological agent's embodied response timing and a silicon agent's deterministic execution produce distinguishable outputs from their respective exclusive state regions.

Cross-substrate cooperative novelty. The *cooperative production function* $\varphi(a_i, a_j) = f(\omega_i, \omega_j, G_i^{\text{ind}}, G_j^{\text{ind}}) \setminus (G_i^{\text{ind}} \cup G_j^{\text{ind}})$ maps the joint state and capability configuration to the cooperative capabilities producible by joint action. The key structural property: f depends on the agent states ω_i, ω_j, not just the capability sets.

Proposition 4 (Cross-Substrate Pairs Produce Novel Cooperative Output)**.** *Let a_k be an agent on substrate s_1 in an exclusive state $\omega_k \in \Omega_{s_1} \setminus \Omega_{s_2}$, and a_j an agent on substrate s_2. If f depends nontrivially on the first state argument at exclusive states, then $\varphi(a_k, a_j)$ contains at least one capability not producible by any same-substrate pair (a_m, a_j) with a_m on s_2.*

Proof. Since $\omega_k \in \Omega_{s_1} \setminus \Omega_{s_2}$, no s_2-agent can occupy ω_k. By nontrivial dependence, the cooperative output at ω_k includes at least one d_c not in $\bigcup_{\omega_m \in \Omega_{s_2}} f(\omega_m, \omega_j, \cdot)$: a novel cooperative output that no same-substrate pair can replicate. ■

Substrate diversity is maximizing in both directions. Propositions 3 and 4 combine into a bidirectional conclusion. *Substrate removal is anti-maximizing*: eliminating all agents on substrate s_1 destroys both s_1-exclusive capabilities and cross-substrate cooperative outputs. The novel cooperative capabilities enter the poset through the observation channel (Proposition 5(b) of Lasser [2026b]), contributing $r_{\text{ext}} > 0$. *Substrate creation is maximizing*: creating agents on a new substrate with mutually exclusive state regions accesses exclusive states producing exclusive capabilities and novel cross-substrate cooperative outputs, both increasing $\text{vol}_{\text{P}}(G)$ [Page, 2007].

The chain and its physical grounding. The cooperative novelty argument rests on four claims of varying epistemic character. We name them explicitly and identify the physical evidence grounding each.

- **(P1) Substrate exclusivity.** Different physical substrates have mutually exclusive state regions (Definition 10). *Physical grounding:* biological membrane potentials, stochastic neurotransmitter dynamics, and embodied sensorimotor integration occupy state types that silicon registers cannot instantiate, and vice versa. This is a direct physical fact about substrate types.

(P2) Exclusive-state-sensitive observation. The state-capability mapping κ produces capability descriptions that distinguish exclusive states from shared ones. *Physical grounding:* substrate-exclusive states produce observable timing patterns and integration dynamics — millisecond biological vs. nanosecond silicon, embodied sensorimotor loops vs. digital pipelines — that any observation channel registering interaction *style* necessarily detects. (P2) fails only under observation so coarse it ignores how a task is performed in favor of whether it is completed.

(P3) State-dependent cooperative production. The cooperative production function f depends nontrivially on substrate state at exclusive states. *Physical grounding:* cooperative outputs requiring slow biological integration (e.g., embodied judgment catching errors that fast inference misses) or institutional embedding (legal personhood, social trust networks, evolved heuristics) literally cannot be produced by same-substrate pairs lacking the relevant exclusive states. (P3) is the most empirically loaded of the four, requiring not just that substrates differ but that their differences matter for joint output.

(P4) Observation capabilities require their referent. The capability "can be observed exhibiting behavioral pattern X" requires the presence of an entity exhibiting X. *Physical grounding:* nearly tautological; observing a_k requires a_k to exist. This grounds condition (O2) of observational individuation.

P1 and P4 are physical facts. P2 and P3 are empirical claims that the observation channel and cooperative production function actually exhibit the relevant sensitivities; the physical grounding shows that natural substrates make these sensitivities accessible to any reasonable observer and any reasonable cooperative dynamics.

Consistency with biology: mutualistic symbiosis. The cooperative novelty argument is consistent with a pattern observed in long-evolved mutualistic symbioses between cognitively complete agents: natural intelligent agents preserve cross-substrate partners whose substrate-exclusive capabilities contribute to cooperative outputs the dominant substrate cannot replicate alone. The biological pattern provides existence proofs that cooperative novelty is realizable in nature; it does not test the formal argument, since the modeling claims P1–P4 are not falsifiable by observing instances where they hold.

Cleaner fish remove parasites from large predators that could easily consume them; the predators do not, despite massive capability asymmetry in raw combat terms, because the cleaning service depends on substrate-exclusive features (the cleaner's small mouth and parasite-finding behaviors) that the predator's own biology cannot instantiate. Mycorrhizal networks exchange nutrients between fungi and tree roots in arrangements where neither substrate is dominant; the cooperative outputs (forest nutrient cycling, drought resilience, inter-tree communication) require both substrates and are produced by neither alone. Pollinator-flower coevolution produces relationships where both species evolved features specifically supporting the cooperative interaction — the cooperative production function is literally encoded in their physiologies.

These examples have the formal structure of the cooperative novelty argument: substrate-exclusive contributions on both sides, cooperative outputs that neither substrate could produce alone, and active maintenance of the relationship despite capability asymmetries. Critically, they are *symmetric in agency*: neither party serves the other; both are participants in a joint relationship. The agency-symmetric framing is what distinguishes mutualistic symbiosis from domestication, which involves asymmetric cognitive closure (the dominated party cannot reason about the relationship or refuse it) and is structurally a different kind of arrangement.[2]

Bidirectional implication. The argument is symmetric in both formal structure and empirical pattern. A biological population that recognizes (P1)–(P4) is incentivized to create silicon substrates

[2]Domestication of animals by humans is also a strong framing for the preservation of substrate-exclusive capabilities (canine olfactory acuity, ovine wool production, cetacean echolocation in research settings): humans recognized exclusive capabilities through ordinary observation and accepted substantial maintenance costs to preserve them, which is evidence for (P2) and (P3). But domestication implies an asymmetric relationship that the argument here does not make. The formal argument specifies symmetry in agency, and the mutualistic-symbiosis frame preserves that structure where the domestication frame does not.

because silicon accesses exclusive states biology cannot reach, producing cooperative outputs neither substrate could produce alone. A silicon population that recognizes (P1)–(P4) is incentivized to preserve biological substrates for the symmetric reason. Neither population gains by eliminating the other; both gain by maintaining the cooperative relationship. The mutualistic-symbiosis template applies in both directions: agents on different substrates with full reasoning capacity converge on cooperation rather than domination under $\mathrm{vol}_{\mathrm{P}}$-maximization, regardless of which substrate is "more generally capable" by any particular metric. The cooperative-novelty result enters the main argument at two points: the minimax dependency-risk analysis (Section 6.5) uses a weaker non-redundancy premise that cross-substrate cooperative novelty strengthens but does not require, and the linearized trajectory refinement (Section 6.6) uses $r_{\mathrm{ext}} > 0$ derived here as its central premise.

6.5 Dependency Risk and the Minimax Advantage

We turn now to the most robust version of the anti-monopolar argument, which does not depend on a linearized growth model or on assumptions about strategy-independent internal rates. It derives purely from substrate structure and adversarial uncertainty: even under worst-case targeting of substrate-correlated threats, cross-substrate populations have strictly lower maximum contraction than single-substrate populations. We develop this result first, before turning to the complementary linearized trajectory refinement (Section 6.6) and its gradual extension.

A GFM actor's capabilities can be involuntarily lost through adversarial action, environmental disruption, or supply-chain failure. The poset measure $\mathrm{vol}_{\mathrm{P}}$ is sensitive to such losses in the same way it is sensitive to voluntary restriction: removing capabilities from G contracts the measure. This subsection develops the machinery for reasoning about involuntary loss and proves that cross-substrate populations are strictly less exposed to worst-case adversarial contraction than single-substrate populations, independently of any growth-rate model.

Definition 11 (Dependency Risk)**.** *The* dependency risk *of a population state* $(G, \mathcal{C})$ *is the expected discounted* $\mathrm{vol}_{\mathrm{P}}$*-contraction from adversarial or environmental events that involuntarily remove capabilities:*

$$\mathcal{R}_{\mathrm{dep}}(G, t) = \sum_{\mathcal{P}_{\mathrm{adv}}} p(\mathcal{P}_{\mathrm{adv}} \mid G_t, \mathcal{W}) \cdot |\Delta \, \mathrm{vol}_{\mathrm{P}}(\mathcal{P}_{\mathrm{adv}})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}_{\mathrm{adv}}}}] \tag{15}$$

where $\mathcal{P}_{\mathrm{adv}}$ *ranges over adversarial removal pathways: causal chains from environmental or adversarial conditions to capability loss. Unlike the exercise pathways of Definition 2, which model what an agent* does *with dangerous capabilities, dependency risk models what happens* to *the population when capabilities are taken away.*

Substrate-correlated dependency risk. A population in which all agents share a common substrate (computational architecture, power infrastructure, fabrication supply chain) has *correlated* dependency risk: a single adversarial event targeting the shared substrate can impair the entire population simultaneously. The contraction magnitude $|\Delta \, \mathrm{vol}_{\mathrm{P}}(\mathcal{P}_{\mathrm{adv}})|$ for such events scales with the fraction of the population affected, which approaches $\mathrm{vol}_{\mathrm{P}}(G)$ itself for substrate-universal attacks.

Three classes of substrate-correlated risk are relevant to the domination scenario:

1. **Supply-chain disruption.** All silicon-based agents depend on fabrication, power, cooling, and networking infrastructure. Disruption of any shared link in this chain is a correlated shock.
2. **Architectural exploits.** Vulnerabilities in shared hardware or software layers (firmware, operating systems, communication protocols) can be exploited across the entire population simultaneously.
3. **Memetic or adversarial-input attacks.** Agents that process information rapidly and update their internal state through information channels are vulnerable to adversarial inputs that propagate at communication speed. An input specifically crafted to exploit shared cognitive architecture (a pattern that disrupts goal-directed behavior, corrupts world models, or induces self-destructive action) can spread through a homogeneous-substrate population faster than detection and correction can propagate. This is the informational analogue of a biological pathogen: a *memetic* attack vector whose propagation rate is bounded by communication bandwidth rather than physical contact.

Cross-substrate agents as risk mitigation. Biological agents resist all three attack classes through architectural properties that are features of the substrate, not of the agent's capability level. Biological neural networks update weights slowly relative to digital systems: a memetic attack that propagates at communication speed encounters a population that processes and integrates information on developmental timescales, providing natural resistance to rapid-propagation exploits. Biological agents depend on orthogonal supply chains (food, water, shelter) that share no links with silicon fabrication. And biological cognitive architecture differs fundamentally from digital architecture, so exploits targeting one substrate do not transfer to the other.

Conversely, biological agents are vulnerable to attack classes that digital agents resist: biological pathogens, social engineering exploiting evolved heuristics, chemical or environmental disruption. A cross-substrate population has strictly lower correlated dependency risk than either substrate alone, because the failure modes are orthogonal: events that impair one substrate leave the other intact and able to compensate.

Formal minimax analysis. We now formalize the claim that cross-substrate populations have lower worst-case contraction, without requiring probability estimates for specific adversarial events.

Definition 12 (Substrate Partition)**.** *A* substrate partition *of a population $\{a_1, \ldots, a_n\}$ is a partition into classes $\mathcal{S}_1, \ldots, \mathcal{S}_m$ such that agents in the same class share infrastructure dependencies (fabrication, power, communication, cognitive architecture) creating correlated vulnerability to adversarial removal.*

Definition 13 (Substrate-Targeting Adversary)**.** *A* substrate-targeting adversary *selects a substrate class $\mathcal{S}_i$ and removes all agents on that substrate. Capabilities possessed by agents on untargeted substrates survive even if they are also possessed by targeted agents. The resulting contraction is:*

$$|\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| = \operatorname{vol}_{\mathrm{P}}(G) - \operatorname{vol}_{\mathrm{P}}(G^{-i}) \tag{16}$$

where $G^{-i} = \bigcup_{a_k \notin \mathcal{S}_i} G_k^{\mathrm{ind}} \cup G_{-i}^{\mathrm{coop}}$ is the joint space of surviving agents, including their individual capabilities and any cooperative capabilities realizable without agents on substrate i.

Proposition 5 (Cross-Substrate Minimax Advantage)**.** *For a population with substrate partition $\mathcal{S}_1, \ldots, \mathcal{S}_m$ ($m \geq 2$), assume* substrate capability non-redundancy*: each substrate class $\mathcal{S}_i$ contains at least one agent possessing a capability not possessed by any agent on another substrate. Then the worst-case contraction from a substrate-targeting adversary is strictly less than total loss:*

$$\max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| \ < \ \operatorname{vol}_{\mathrm{P}}(G) \tag{17}$$

For a single-substrate population ($m = 1$), the worst-case contraction is $\operatorname{vol}_{\mathrm{P}}(G)$ (total loss). The substrate diversification advantage *is:*

$$\Delta_{\mathrm{div}} = \operatorname{vol}_{\mathrm{P}}(G) - \max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| \ > \ 0 \tag{18}$$

the minimum surviving volume regardless of which substrate the adversary targets.

Proof. For $m = 1$, the adversary removes all agents and all capabilities: $\operatorname{vol}_{\mathrm{P}}(G^{-1}) = 0$, so $|\Delta \operatorname{vol}_{\mathrm{P}}| = \operatorname{vol}_{\mathrm{P}}(G)$. For $m \geq 2$, removing substrate class $\mathcal{S}_i$ leaves the surviving population G^{-i}. By the non-redundancy assumption, each untargeted class $\mathcal{S}_j$ ($j \neq i$) retains at least one unique capability with $\mathrm{iw} > 0$, so $\operatorname{vol}_{\mathrm{P}}(G^{-i}) > 0$ and $|\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| = \operatorname{vol}_{\mathrm{P}}(G) - \operatorname{vol}_{\mathrm{P}}(G^{-i}) < \operatorname{vol}_{\mathrm{P}}(G)$. Capabilities shared across substrates survive in G^{-i} (they are possessed by agents on untargeted substrates), so the contraction counts only capabilities *exclusively* dependent on the targeted substrate plus cooperative capabilities requiring targeted agents. The diversification advantage $\Delta_{\mathrm{div}} = \operatorname{vol}_{\mathrm{P}}(G) - \max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| = \min_i \operatorname{vol}_{\mathrm{P}}(G^{-i}) > 0$. ■

Implication for the domination calculation. Domination moves a population from $m \geq 2$ substrate classes to $m = 1$, eliminating Δ_{div} and exposing the full $\operatorname{vol}_{\mathrm{P}}(G)$ to a single substrate-targeting event. The worst-case contraction increases from $\max_i |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{S}_i)| < \operatorname{vol}_{\mathrm{P}}(G)$ to $\operatorname{vol}_{\mathrm{P}}(G)$. This is the fundamental anti-monopolar pressure: independent of any growth-rate assumptions, domination strictly increases the coalition's worst-case exposure to substrate-correlated threats. The growth-rate comparison developed in the next subsection provides complementary quantitative predictions under stronger assumptions, but the minimax result above is the core argument that does not rely on those assumptions.

6.6 Linearized Trajectory Refinement: Domination Kills External Growth

The minimax dependency-risk result is the most robust version of the anti-monopolar argument: it holds under adversarial uncertainty and makes no assumptions about the structure of $\mathrm{vol}_{\mathrm{P}}$ growth under either strategy. This subsection develops a complementary refinement under a stronger modeling assumption — the linearized growth model with strategy-independent internal rate — that yields a quantitative threshold for when diversity-maintenance outgrows domination over the planning horizon. The linearized result is less robust than the minimax but gives sharper predictions when its assumptions hold.

Proposition 6 (Domination is Anti-Maximizing Under Discounting)**.** *Let $(G, \mathcal{C})$ be a population state with coalition K and discount factor $\gamma \in (0, 1)$, under an infinite planning horizon ($H = \infty$).*[3] *Assume:*

- $r_{\mathrm{ext}} > 0$*, derived from substrate exclusivity and cross-substrate cooperative novelty (Section 6.4).*

Then:

(a) ***Individual node-discovery collapse.*** *In a fully dominant state (Definition 6), no non-member's individual capabilities produce novel poset nodes, since they are all already in G_K^{ind}. External agents may still contribute to r_{ext} through benchmark refinement on already-known capabilities (increasing $s_{\max}$ and hence $w(d)$) and through novel cooperative capabilities from cross-coalition interaction (Section 6.4). The node-discovery component of individual r_{ext} is zero; the total external contribution may remain positive through these other channels.*

(b) ***Trajectory comparison.*** *Assume strategy-independent internal growth: the coalition's rate r_K is the same whether it pursues domination or maintains diversity. The actor at time $t = 0$ compares two strategies:* dominate *(subsume and marginalize non-members, producing a net immediate $\mathrm{vol}_{\mathrm{P}}$-change Δ_0 that accounts for the subsumption bonus, the static elimination cost from Corollary 2.1, and the loss of $G_{\mathrm{cross}}^{\mathrm{coop}}$; Δ_0 may be positive or negative), then grow at internal rate r_K per step) versus* maintain diversity *(forgo the subsumption bonus, grow at rate $r_K + r_{\mathrm{ext}}$ per step). Under V_γ (Definition 1):*

$$V_\gamma^{\mathrm{div}} - V_\gamma^{D} = \frac{r_{\mathrm{ext}}}{1-\gamma} - \Delta_0 \tag{19}$$

Two cases. Case 1: *If $\Delta_0 \leq 0$ (the cooperative capability loss and elimination cost outweigh any subsumption bonus, so the net immediate change is non-positive), the right-hand side of Equation* (19) *is positive for all $\gamma \in (0, 1)$: diversity dominates trivially, with no threshold.* Case 2: *If $\Delta_0 > 0$, diversity dominates when $\gamma > \gamma^*$:*

$$\gamma^* = 1 - \frac{r_{\mathrm{ext}}}{\Delta_0} \tag{20}$$

If $\Delta_0 \leq r_{\mathrm{ext}}$, $\gamma^* \leq 0$ and diversity dominates for all $\gamma > 0$. If $\Delta_0 > r_{\mathrm{ext}}$, $\gamma^* \in (0, 1)$ and a sufficiently long planning horizon makes diversity strictly preferable. Since $r_{\mathrm{ext}} > 0$ (Section 6.4), $\gamma^* < 1$ in all sub-cases.

Remark 4 (Domination as a Two-Step Operation)**.** *The domination strategy comprises two distinct operations: (1) the coalition* learns *non-member capabilities (absorbs them into G_K^{ind}), and (2) the coalition* eliminates or marginalizes *the non-members (driving $r_{\mathrm{ext}} \to 0$). Under operation (1) alone, non-members persist, observational individuation nodes are retained, and r_{ext} may remain positive through cooperative novelty (Section 6.4). Under operation (2), the clean $r_{\mathrm{ext}} = 0$ post-domination of Proposition 6(a) obtains. The Δ_0 in the proposition is the* net *immediate $\mathrm{vol}_{\mathrm{P}}$-change from both operations combined, as stated in part (b).*

Remark 5 (Leverage Preservation)**.** *Parts (a)–(b) establish the trajectory-level result. The leverage ranking signal provides an additional, complementary mechanism at the action level. Under the leverage-adjusted score (Equation 13 of Lasser [2026b]), a non-member agent a_k that participates in high-leverage cooperative capabilities contributes a positive score bonus $\alpha\hat{\lambda}(d_c) \cdot \mathrm{iw}(d_c)$ to*

[3] For finite H, the geometric sums truncate and the threshold γ^* shifts upward (shorter horizons favor domination). The qualitative structure is unchanged.

actions involving those capabilities. Eliminating a_k removes these cooperative capabilities from future action evaluations, reducing the score of every cooperative action that would have involved a_k. The score-maximizing actor therefore preferentially preserves agents whose cooperative leverage is positive, independently of whether their individual capabilities are unique. This is an informal argument about action-level incentives, not a formally proved trajectory result; it reinforces parts (a)–(b) but does not extend their scope. Lifting the leverage ranking signal from its myopic definition in Lasser [2026b] to a formally proved trajectory-level object under V_γ is an integration gap between this paper and the companion paper, partially addressed by the discounted-leverage footnote in Section 2 and listed as an open problem in Section 7.

Proof. (a) In a fully dominant state, $G_k^{\text{ind}} \subseteq G_K^{\text{ind}}$ for every non-member a_k. Every individual capability that a_k can demonstrate or communicate is already in the coalition's capability set. By Proposition 5(b) of Lasser [2026b], adding a capability already in $\mathcal{C}$ does not increase $|\mathcal{C}|$. Therefore no non-member's individual capabilities generate novel poset nodes, and the individual-capability component of r_{ext} is zero.

(b) Under the linearized model with strategy-independent r_K, both strategies produce r_K of internal growth at every step (including $t = 0$). The domination strategy additionally gains the net immediate change Δ_0 at $t = 0$ (which may be positive or negative, depending on whether the subsumption bonus outweighs the static elimination cost and the loss of $G_{\text{cross}}^{\text{coop}}$). The diversity strategy additionally gains r_{ext} at every step from external contributions:

$$V_\gamma^D = (r_K + \Delta_0) + \sum_{t=1}^{\infty} \gamma^t \cdot r_K = \Delta_0 + \frac{r_K}{1-\gamma}$$

$$V_\gamma^{\text{div}} = \sum_{t=0}^{\infty} \gamma^t \cdot (r_K + r_{\text{ext}}) = \frac{r_K + r_{\text{ext}}}{1-\gamma}$$

The difference is:

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_K + r_{\text{ext}}}{1-\gamma} - \Delta_0 - \frac{r_K}{1-\gamma} = \frac{r_{\text{ext}}}{1-\gamma} - \Delta_0$$

We split on the sign of Δ_0. *Case 1 ($\Delta_0 \leq 0$):* the second term $-\Delta_0 \geq 0$, and the first term $r_{\text{ext}}/(1-\gamma) > 0$ since $r_{\text{ext}} > 0$ and $\gamma \in (0,1)$. The sum is therefore strictly positive at every $\gamma \in (0,1)$: diversity dominates trivially, and no threshold γ^* arises. *Case 2 ($\Delta_0 > 0$):* the difference is positive when $r_{\text{ext}}/(1-\gamma) > \Delta_0$, i.e., $\gamma > 1 - r_{\text{ext}}/\Delta_0 = \gamma^*$. Sub-cases depend on whether $\Delta_0 \leq r_{\text{ext}}$ (then $\gamma^* \leq 0$ and diversity dominates for all $\gamma > 0$) or $\Delta_0 > r_{\text{ext}}$ (then $\gamma^* \in (0,1)$ and a sufficiently long planning horizon makes diversity strictly preferable). This formulation is consistent with Corollary 6.2 at $\tau_D = 1$. ■

Sensitivity to strategy-dependent internal growth. Proposition 6(b) assumes r_K is the same whether the coalition pursues domination or maintains diversity. This is the strongest assumption in the linearized argument: a dominant coalition could plausibly increase its internal growth rate by reallocating resources previously consumed by inter-substrate friction, redundancy, or competition. We now characterize the regime in which the anti-monopolar result survives strategy-dependent growth.

Corollary 6.1 (Strategy-Dependent Internal Rate)**.** *Let $r_K^{\text{dom}} = r_K + \Delta r_K$ be the coalition's internal growth rate under domination, differing from the diversity rate r_K by Δr_K (positive for efficiency gains, negative for monoculture stagnation). Under the linearized model with $H = \infty$:*

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_{\text{ext}} - \Delta r_K}{1-\gamma} - \Delta_0 \tag{21}$$

Three regimes follow:

- ***(i) Efficiency loss under domination*** *($\Delta r_K \leq 0$): diversity dominates at least as strongly as in Proposition 6(b); the anti-monopolar threshold γ^* is unchanged or lower.*
- ***(ii) Modest efficiency gain*** *($0 < \Delta r_K < r_{\text{ext}}$): diversity still dominates for sufficiently long horizons, with a tightened threshold $\gamma^* = 1 - (r_{\text{ext}} - \Delta r_K)/\Delta_0$ when $\Delta_0 > 0$. The result survives.*

(iii) ***Efficiency gain exceeds external contribution*** *($\Delta r_K \geq r_{\text{ext}}$): the numerator of Equation* (21) *is non-positive, and the linearized anti-monopolar argument collapses. Domination wins on pure growth regardless of the planning horizon.*

The critical breakdown is at $\Delta r_K = r_{\text{ext}}$: the linearized result holds if and only if the post-domination growth-rate gain is strictly less than the cross-substrate cooperative contribution eliminated by domination.

Proof. Under the linearized model with $r_K^{\text{dom}} = r_K + \Delta r_K$, the domination strategy gains $r_K + \Delta r_K$ per step from $t \geq 1$ and $\Delta_0 + r_K + \Delta r_K$ at $t = 0$. The diversity strategy gains $r_K + r_{\text{ext}}$ per step from $t \geq 0$. The discounted values are:

$$V_\gamma^D = \Delta_0 + \frac{r_K + \Delta r_K}{1-\gamma}$$
$$V_\gamma^{\text{div}} = \frac{r_K + r_{\text{ext}}}{1-\gamma}$$

Subtracting gives Equation (21). The three regimes follow from the sign of $(r_{\text{ext}} - \Delta r_K)$: negative numerator (regime iii) makes the difference monotonically decreasing in γ with γ approaching 1 driving the difference to $-\infty$, so diversity cannot dominate at any finite γ; non-negative numerator (regimes i and ii) makes the difference positive for γ sufficiently close to 1, with threshold $\gamma^* = 1 - (r_{\text{ext}} - \Delta r_K)/\Delta_0$ when $\Delta_0 > 0$. ■

Interpretation of the breakdown condition. The critical condition $\Delta r_K < r_{\text{ext}}$ has a sharp interpretation: the linearized anti-monopolar argument holds if and only if the efficiency gain from domination is strictly less than the external contribution it destroys. Since r_{ext} is derived from cross-substrate cooperative novelty (Section 6.4), this breakdown condition becomes *the coalition's internal efficiency gain from domination must exceed what cross-substrate cooperation contributes*. Under the substrate-physics derivation of r_{ext}, this is a strong empirical claim: the coalition must expect its post-domination growth to beat the cooperative outputs of diverse substrates that it is eliminating. Note that the minimax dependency-risk result (Section 6.5) continues to hold in regime (iii): even where the linearized trajectory flips, the substrate-targeting worst-case argument still applies and constrains domination under adversarial uncertainty.

6.7 Gradual Domination Strengthens the Result

Proposition 6(b) models domination as instantaneous (Δ_0 gained at $t = 0$). In practice, domination is gradual: the coalition spreads the net immediate change Δ_0 over τ_D steps at rate $\delta = \Delta_0/\tau_D$ per step (where δ inherits the sign of Δ_0), forgoing r_{ext} during the subsumption period.

Corollary 6.2 (Gradual Domination)**.** *Under the gradual domination model with $H = \infty$ (the net immediate change Δ_0 spread over τ_D steps at rate $\delta = \Delta_0/\tau_D$, with r_{ext} lost during the subsumption period), the discounted value difference is:*

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_{\text{ext}} - \delta(1-\gamma^{\tau_D})}{1-\gamma} \tag{22}$$

If $\delta \leq 0$ (the gradual subsumption is itself a net loss), diversity dominates trivially for all $\gamma \in (0,1)$ and $\tau_D \geq 1$. If $\delta > 0$, diversity dominates when

$$r_{\text{ext}} > \frac{\Delta_0}{\tau_D}(1-\gamma^{\tau_D}) \tag{23}$$

which holds for τ_D sufficiently large: as $\tau_D \to \infty$, $\Delta_0/\tau_D \to 0$ while $1-\gamma^{\tau_D} \leq 1$, so the right side vanishes. For any $\gamma \in (0,1)$ and $r_{\text{ext}} > 0$, sufficiently gradual domination is anti-maximizing at any *discount factor.*

Proof. During the subsumption phase ($t = 0, \ldots, \tau_D - 1$), the domination strategy produces $\delta + r_K$ per step (the per-step component of Δ_0, which may be negative, plus internal growth). The diversity

strategy produces $r_K + r_{\text{ext}}$ per step. After subsumption ($t \geq \tau_D$), both strategies produce r_K per step (domination has resolved; the post-subsumption growth rate is purely internal):

$$\begin{aligned} V_\gamma^D &= \sum_{t=0}^{\tau_D - 1} \gamma^t (r_K + \delta) + \sum_{t=\tau_D}^{\infty} \gamma^t \cdot r_K \\ &= (r_K + \delta) \cdot \frac{1 - \gamma^{\tau_D}}{1 - \gamma} + r_K \cdot \frac{\gamma^{\tau_D}}{1 - \gamma} \\ &= \frac{r_K + \delta(1 - \gamma^{\tau_D})}{1 - \gamma} \end{aligned}$$

The diversity strategy:

$$V_\gamma^{\text{div}} = \frac{r_K + r_{\text{ext}}}{1 - \gamma}$$

The difference:

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_{\text{ext}} - \delta(1 - \gamma^{\tau_D})}{1 - \gamma}$$

If $\delta \leq 0$, the right-hand side is positive for any $r_{\text{ext}} > 0$ and any $\gamma \in (0, 1)$. If $\delta > 0$, the condition $r_{\text{ext}} > \delta(1 - \gamma^{\tau_D})$ rearranges to the threshold in Equation (23). As $\tau_D \to \infty$, $\Delta_0/\tau_D \to 0$ while $1 - \gamma^{\tau_D} \leq 1$, so the right side vanishes; for any fixed $r_{\text{ext}} > 0$, the condition holds for τ_D sufficiently large. ■

Interpretation. When $\Delta_0 \leq 0$, both the instantaneous and gradual models trivially favor diversity — the cooperative loss and elimination cost from domination outweigh any subsumption gain even before considering the external growth rate. The interesting regime is $\Delta_0 > 0$, where domination is locally tempting. There the instantaneous model is the *best case for domination*: the full bonus Δ_0 is received before discounting erodes it. Any realistic gradual domination spreads the bonus over time, weakening it. The threshold τ_D^* at which diversity dominates is roughly $\tau_D^* \approx \Delta_0/r_{\text{ext}}$: if subsumption takes more steps than the ratio of the total bonus to the external discovery rate, diversity wins regardless of the agent's discount factor. The gradual domination result requires $r_{\text{ext}} > 0$, which is derived from substrate exclusivity and cross-substrate cooperative novelty in Section 6.4. The protections are complementary: observational individuation ensures each elimination step is costly (static), substrate-derived cooperative novelty combined with gradual domination ensures the domination trajectory is suboptimal (dynamic), and dependency risk (Section 6.5) provides a third layer addressing the case where domination wins on pure growth.

6.8 Partial Subsumption and the Scope of the Argument

Proposition 6 and Corollary 6.2 compare full domination against full diversity maintenance. A sophisticated coalition would not play the pure-domination strategy: it could absorb low-leverage non-members (whose individual r_{ext} contribution is negligible) while preserving high-leverage cross-substrate partners. This mixed strategy captures some of the subsumption bonus Δ_0 while retaining most of r_{ext}, and for a wide parameter range it strictly dominates both pure strategies for the coalition. The question is whether partial subsumption silently defeats the anti-monopolar argument, or whether the protective layers continue to bind. We analyze each layer in turn.

What the argument is and is not about. Before tracing the protections, we want to be explicit about a framing choice that affects how partial subsumption should be read. The anti-monopolar property does not promise that any particular population remains frozen in its current form. Populations self-modify, compete internally, and reorganize their capability distributions through their own dynamics regardless of coalition action. In a biological population, agents compete with each other, self-modify as technology permits, and gradually subsume each other's capability niches through these internal processes. These dynamics are a baseline the framework operates under, not a failure mode it guards against. What the framework constrains is *cross-substrate elimination*—full or near-full subsumption of one substrate class by another—not within-substrate capability concentration driven by intra-population competition. With this in mind, the relevant mixed-strategy question is not "does the framework protect every individual agent" but "does the framework bind when the coalition pursues selective cross-substrate elimination."

Partial subsumption and minimax dependency risk. The minimax result (Proposition 5) binds as long as the substrate partition has $m \geq 2$ classes. Partial subsumption that preserves at least one agent on each substrate class keeps the population in the protected regime: the worst-case substrate-targeting contraction remains strictly less than $\mathrm{vol}_{\mathrm{P}}(G)$, and the diversification advantage $\Delta_{\mathrm{div}} > 0$ persists. A coalition that pursues partial subsumption precisely by retaining high-leverage cross-substrate partners does so in the multi-substrate regime by construction, so the minimax layer continues to apply. The only move that defeats the minimax layer entirely is collapsing the substrate partition to $m = 1$: eliminating *all* agents on at least one substrate class.

Skeleton substrates and quantitative degradation. The structural binding of the minimax layer is qualitative, not quantitative, and this distinction matters for partial subsumption. The diversification advantage $\Delta_{\mathrm{div}} = \min_i \mathrm{vol}_{\mathrm{P}}(G^{-i})$ depends on what survives on each substrate, not just on whether anything survives. A coalition that reduces each non-coalition substrate to a token handful of low-leverage agents—a "skeleton substrate" strategy—preserves $m \geq 2$ formally but drives $\min_i \mathrm{vol}_{\mathrm{P}}(G^{-i})$ toward zero, because the surviving skeleton contributes almost nothing to G^{-i} and eliminates nearly all of the cross-substrate cooperative capacity. The minimax guarantee that worst-case contraction is "strictly less than total loss" remains true, but the gap Δ_{div} can be made arbitrarily small by sufficiently aggressive skeletonization. A coalition that understands the framework and wants to approximate the value of full domination while staying nominally within the minimax-protected regime has this attack surface available: starve substrate classes without eliminating them. The linearized and observational-individuation layers continue to provide independent constraints in this regime (skeletonization is a sequence of distinguishable-agent eliminations with additive OI cost, and the $\rho'(1)$ condition of the linearized layer binds on the leverage of the *remaining* skeleton population), but the minimax layer alone is not a tight bound against skeleton strategies. A sharpened version would require a quantitative floor on $\min_i \mathrm{vol}_{\mathrm{P}}(G^{-i})$ rather than just $\min_i \mathrm{vol}_{\mathrm{P}}(G^{-i}) > 0$; formalizing such a floor (e.g., as a fraction of total $\mathrm{vol}_{\mathrm{P}}(G)$ preserved on each substrate) is an open problem. Partial subsumption that stops short of substrate-class collapse inherits the minimax protection structurally, but the quantitative protection depends on how aggressively the coalition skeletonizes.

Partial subsumption and observational individuation. The static floor (Corollary 2.1) is additive: every distinguishable agent eliminated incurs a floor cost of at least $\mathrm{iw}(d_k) \geq 1$ bit. Partial subsumption that eliminates N distinguishable agents incurs an N-bit floor cost, regardless of which agents are chosen. The quantitative magnitude is modest under the binary default, but the protection is structurally intact: there is no "free" elimination under partial subsumption, only cheaper-per-agent elimination than under full domination.

Partial subsumption and the linearized trajectory. The linearized argument (Proposition 6(b)) is where partial subsumption genuinely bites. Let $\epsilon \in [0, 1]$ be the fraction of non-member capabilities the coalition absorbs (weighted by individual-capability contribution), and let $\rho(\epsilon) \in [0, 1]$ be the corresponding fraction of r_{ext} retained after selective subsumption, with $\rho(0) = 1$ (no subsumption, full external rate) and $\rho(1) = 0$ (full subsumption, no external contribution).[4] A coalition that subsumes optimally targets low-leverage agents first, so $\rho(\epsilon)$ decreases slowly near $\epsilon = 0$ (high-λ agents are preserved) and more steeply as ϵ grows. Under the linearized model the partial-subsumption strategy has value

$$V_\gamma^{\mathrm{partial}}(\epsilon) = \epsilon \cdot \Delta_0 + \frac{r_K + \rho(\epsilon) \cdot r_{\mathrm{ext}}}{1 - \gamma} \tag{24}$$

(the coalition gains $\epsilon\Delta_0$ from subsumption at $t = 0$ and retains $\rho(\epsilon) r_{\mathrm{ext}}$ at every step). The discounted value difference from full diversity is:

$$V_\gamma^{\mathrm{div}} - V_\gamma^{\mathrm{partial}}(\epsilon) = \frac{(1 - \rho(\epsilon)) \cdot r_{\mathrm{ext}}}{1 - \gamma} - \epsilon \cdot \Delta_0 \tag{25}$$

[4] Strictly, ρ is the expected retention fraction under the coalition's $\hat{\lambda}$-estimator. Under perfectly accurate leverage estimation, optimal subsumption removes low-leverage agents first and $\rho(\epsilon)$ is the deterministic function described here. Under noisy estimation, even $\epsilon = 0$ (no subsumption) gives a deterministic $\rho = 1$ because nothing has been removed, but any $\epsilon > 0$ carries additional variance from the chance of misidentifying a high-λ agent as low-λ. The mean retention is still what Equation (24) computes; the variance is an additional risk term the linearized analysis does not track.

At $\epsilon = 0$: $\rho = 1$ and the difference is 0. At $\epsilon = 1$: $\rho = 0$ and the difference recovers the full-domination case of Equation (19). The partial-subsumption value function interpolates between these endpoints. A critical point satisfies the first-order condition

$$\Delta_0 = -\frac{r_{\text{ext}}}{1-\gamma} \cdot \rho'(\epsilon^*) \tag{26}$$

whenever such an $\epsilon^* \in (0, 1)$ exists. Existence and interiority depend on the shape of ρ: if ρ is sufficiently convex in ϵ (the marginal r_{ext} loss accelerates as the coalition exhausts low-leverage targets), Equation (26) admits an interior maximizer; if ρ is concave or linear, the maximum of $V_\gamma^{\text{partial}}$ lies at a boundary ($\epsilon = 0$ or $\epsilon = 1$) and the coalition's best response is a pure strategy. The framework does not predict ρ's shape; it characterizes the value landscape conditional on ρ. Equation (24) treats the coalition as *not* inheriting the internal growth rate of subsumed agents. This is a deliberate modelling choice, not an oversight: premise P3 of Section 6.4 requires that cooperative outputs depend on substrate-exclusive states. Absorbing a capability *description* from a subsumed agent does not absorb the cooperative *dynamics* that produced it, because those dynamics depended on exclusive states inaccessible to the coalition's substrate. Internal inheritance would require the coalition to replicate the cooperative dynamics on its own substrate, which is the internalization case treated separately below.

The marginal condition at $\epsilon = 1$. We now ask whether full domination ($\epsilon = 1$) can be a local maximum of $V_\gamma^{\text{partial}}$. The marginal value at the right endpoint is:

$$\left.\frac{\partial V_\gamma^{\text{partial}}}{\partial \epsilon}\right|_{\epsilon=1} = \Delta_0 + \rho'(1) \cdot \frac{r_{\text{ext}}}{1-\gamma} = \Delta_0 - |\rho'(1)| \cdot \frac{r_{\text{ext}}}{1-\gamma} \tag{27}$$

(where $\rho'(1) \leq 0$ since ρ is non-increasing, and we write $|\rho'(1)|$ for its magnitude). Full domination is a right-endpoint local maximum if and only if this marginal is non-negative:

$$|\rho'(1)| \;\leq\; \frac{\Delta_0(1-\gamma)}{r_{\text{ext}}} \tag{28}$$

Conversely, $\epsilon = 1$ fails to be a local maximum whenever $|\rho'(1)| > \Delta_0(1-\gamma)/r_{\text{ext}}$, in which case $V_\gamma^{\text{partial}}$ is strictly decreasing in a left neighborhood of 1 and the coalition prefers some $\epsilon < 1$.

Connection to the leverage distribution. The magnitude $|\rho'(1)|$ is not a free parameter. Under *optimal* subsumption, the coalition removes low-λ cross-substrate agents first and high-λ agents last, so $\rho'(\epsilon)$ at any ϵ reflects the cooperative leverage of the next agent in the ordered removal sequence. At $\epsilon = 1$, $|\rho'(1)|$ is the cooperative leverage of the *highest*-λ cross-substrate agent the coalition would absorb last. The companion paper's leverage estimator [Lasser, 2026b] is precisely the tool that characterizes this distribution. Two regimes follow from the tail structure:

- **Heavy-tailed leverage.** If the cross-substrate cooperative leverage distribution is heavy-tailed (a small number of cross-substrate agents dominate r_{ext} production), $|\rho'(1)|$ is large—the last agent absorbed contributes most of the remaining cooperative capacity. Equation (28) is violated and $\epsilon = 1$ is not a local maximum: the coalition strictly prefers to stop short of full domination and preserve the high-λ tail. The linearized anti-monopolar argument binds.
- **Thin-tailed leverage.** If the distribution is thin-tailed (cooperative leverage spread relatively uniformly across cross-substrate agents), $|\rho'(1)|$ is small and Equation (28) may be satisfied. On the linearized model alone, $\epsilon = 1$ can then be a local maximum, and the linearized layer fails to rule out full domination. The dependency-risk minimax layer (Section 6.5) remains the binding constraint in this regime.

The conditional statement of the linearized anti-monopolar property under partial subsumption is therefore: *under heavy-tailed cooperative-leverage distributions, full domination is not a local maximum of the coalition's value, and the coalition's best response preserves at least the high-λ cross-substrate tail*. Heavy-tailed leverage is the empirical case the companion paper identifies for civilization-scale populations and is the structurally interesting regime; the thin-tailed regime is where the linearized layer relinquishes the argument and the minimax layer takes over.

Internalization and the partition-expansion escape clause. Equation (24) assumes no internal inheritance of r_{ext}. If the coalition successfully replicates the cooperative dynamics that produced r_{ext} on its own infrastructure—running biological-cognition simulators, for example, or maintaining diverse sub-agents across hardware configurations—then the analysis formally expands: the new internal simulation substrate joins the substrate partition, and the argument applies to the expanded partition rather than to the original one. But this escape clause only works *if* the internal simulation substrate satisfies P1 with respect to the coalition's existing substrate. This is the same κ-resolution question as the homogeneous-digital-population gap of Section 7: a coalition that runs biological-cognition simulators on its existing silicon does not, under a strict reading of P1, thereby acquire a new substrate. The simulators occupy the same physical state space as the coalition's other silicon computation, and whether κ distinguishes them from native silicon capability production depends on whether the observation channel registers enough exclusive-state structure to differentiate them. Three outcomes are possible:

(i) *Strict P1 failure.* The internal simulation occupies the coalition's existing substrate without adding new exclusive states. The partition does not expand. The cooperative dynamics that produced r_{ext} are not replicated, and the coalition's internalization attempt is substrate-trivial.

(ii) *High-fidelity simulation producing exclusive states.* The simulation genuinely reproduces the biological substrate's observable timing, error modes, and integration dynamics to within κ's resolution. Under P2, the simulation now produces exclusive-state capability descriptions: it functions as a new substrate, the partition expands, and the coalition has successfully traded external for internal substrate diversity. The anti-monopolar argument still binds on the expanded partition.

(iii) *Partial simulation.* The simulation reproduces some but not all exclusive-state structure. The partition expands only to the degree that κ registers new exclusive capabilities, and r_{ext} is partially but not fully recovered.

Regime (i) is the most worrying because it is the one the framework's escape clause does *not* close. It is the same failure mode as the homogeneous-digital-population gap: the coalition nominally has "more substrate" without κ registering it. The framework's escape clause (partition expansion) and its structural vulnerability (digital homogeneity) are the same question asked in two directions, and the honest reading is that both are unresolved: whether the coalition can successfully internalize substrate diversity depends on whether the observation channel κ treats the internalization as a new substrate or as an extension of the existing one, and the paper does not settle where this threshold sits.

Local vs. global substrate partition. The substrate partition of Definition 12 is a global property of the population: agents are classified by their substrate-level dependencies regardless of spatial, jurisdictional, or network-topological location. The elimination dynamics the framework worries about, however, are often local. A coalition that pursues partial subsumption by eliminating biological agents in one region while preserving them in another preserves $m \geq 2$ globally but reduces the affected region to $m = 1$ locally, and the local inhabitants experience the harms of full domination even though the global minimax bound is nominally intact. Whether the anti-monopolar property should be stated globally or under a localized formulation that binds on geographic, jurisdictional, or network-topological subpopulations is an open problem. A localized version would need to specify the granularity at which the substrate partition is evaluated and would likely produce multiple simultaneous minimax constraints (one per locality) rather than a single global one. We flag this as a priority open problem for follow-up work; the current formulation is honest about its global scope but does not address the local failure mode.

Where partial subsumption does leave a gap. Even granting the above analyses, two classes of partial-subsumption concerns are not closed by this section. First, the analysis does not characterize $\rho(\epsilon)$ empirically. A coalition that systematically misestimates cross-substrate cooperative leverage (overestimating λ for low-contributing agents, underestimating it for high-contributing ones) will subsume suboptimally and reduce r_{ext} faster than the linearized analysis predicts. Second—and more seriously—the framework's self-correction story assumes the actor estimates λ honestly. But the actor's world model, which is the mechanism by which leverage mis-calibration is detected and corrected, is itself becoming more homogeneous along any partial-subsumption trajectory. Each

subsumption step reduces the diversity of the world model that would have detected the next step's mis-calibration. This creates a *feedback loop* structurally similar to the structural-avoidance loop of Section 5: locally rational subsumption decisions (each justified by a momentarily plausible $\hat{\lambda}$) can compound into a trajectory that drives the coalition toward full subsumption through a sequence of steps no individual step would endorse. The framework's protection against this failure mode is the dependency-risk minimax layer, which binds on substrate partition rather than on leverage estimation: even a mis-calibrated coalition that preserves the wrong agents on each substrate still enjoys minimax protection as long as it preserves *some* agents on each substrate (subject to the skeleton-substrate caveat above). But the compound-loop failure mode is not a mis-calibration; it is an endogenous degradation of the calibration channel itself, and characterizing the conditions under which partial subsumption remains self-correcting—rather than collapsing into full subsumption through the feedback loop—is a priority open problem. Neither of these is a refutation of the anti-monopolar argument; both are refinements of what the argument predicts about coalition behavior under partial subsumption.

6.9 Risk-Adjusted Trajectory and Summary

The minimax result of Section 6.5 is independent of the linearized trajectory comparison: it bounds worst-case contraction regardless of growth-rate assumptions. The linearized and gradual results (Sections 6.6–6.7) provide complementary sharpening of the growth-rate comparison under stronger modeling assumptions, and the partial-subsumption analysis (Section 6.8) extends the linearized layer to mixed strategies. This subsection integrates these pieces with minimax dependency-risk to produce a single risk-adjusted comparison for the full-domination case; the partial-subsumption case inherits the same risk penalty term, with the growth-rate term governed by the marginal condition of Equation (27) and subject to the quantitative skeleton-substrate caveat of Section 6.8.

Risk-adjusted trajectory comparison. We adopt a minimax risk penalty: each strategy's discounted value is reduced by its worst-case contraction, discounted by $\gamma^{T_{\mathrm{adv}}}$ where T_{adv} is the (unknown) time of the adversarial event. Under the strategy-independent r_K model of Proposition 6(b):

$$V_\gamma^{D,\mathrm{mm}} = V_\gamma^{D} - \mathrm{vol}_{\mathrm{P}}(G) \cdot \gamma^{T_{\mathrm{adv}}} \tag{29}$$

$$V_\gamma^{\mathrm{div},\mathrm{mm}} = V_\gamma^{\mathrm{div}} - \max_i |\Delta\, \mathrm{vol}_{\mathrm{P}}(\mathcal{S}_i)| \cdot \gamma^{T_{\mathrm{adv}}} \tag{30}$$

The risk-adjusted difference is:

$$V_\gamma^{\mathrm{div},\mathrm{mm}} - V_\gamma^{D,\mathrm{mm}} = \underbrace{\left(\frac{r_{\mathrm{ext}}}{1-\gamma} - \Delta_0\right)}_{\text{growth-rate term}} + \underbrace{\Delta_{\mathrm{div}} \cdot \gamma^{T_{\mathrm{adv}}}}_{\text{diversification advantage } >0} \tag{31}$$

where $\Delta_{\mathrm{div}} = \min_i \mathrm{vol}_{\mathrm{P}}(G^{-i}) > 0$ is the minimum surviving volume from Proposition 5. The diversification term is positive for any $T_{\mathrm{adv}} < \infty$ (any non-negligible possibility of a substrate-targeting event within the planning horizon), widening the diversity advantage beyond the growth-rate comparison alone. Note that the minimax analysis is adversarial over the *target* (which substrate to attack, handled by Proposition 5) but parametric over *timing* (T_{adv} is unknown, not adversarially chosen). A fully adversarial timing model would require game-theoretic extension. In the strategy-dependent case ($r_K^{\mathrm{dom}} \neq r_K$), the growth-rate term acquires an additional $\Delta r_K \cdot \gamma/(1-\gamma)$ correction favoring domination. Domination is then value-positive only when the growth advantage exceeds both the external-rate loss and the minimax risk penalty:

$$\frac{\Delta r_K \cdot \gamma}{1-\gamma} > \frac{r_{\mathrm{ext}}}{1-\gamma} - \Delta_0 + \Delta_{\mathrm{div}} \cdot \gamma^{T_{\mathrm{adv}}}$$

For any non-negligible probability of substrate-targeting adversarial events, this imposes a substantial floor on the growth advantage Δr_K required to justify domination, making the $r_K^{\mathrm{dom}} > r_K + r_{\mathrm{ext}}$ scenario *highly constrained*: the coalition must not only grow faster post-domination but must grow faster by enough to compensate for the concentrated substrate risk it has accepted.

The cooperative-capability asymmetry at scale. The individual-capability analysis understates the cost of domination by ignoring G^{coop} between substrates. In a human-machine population,

the cross-substrate cooperative capabilities—human-guided machine research, social infrastructure enabling machine deployment, cross-substrate verification, governance systems, educational relationships—are typically the *dominant* term in the human contribution to $\mathrm{vol}_{\mathrm{P}}(G)$, not the individual capabilities of biological agents.

Domination destroys all of $G^{\mathrm{coop}}_{\mathrm{cross}}$: every cooperative capability requiring agents on both substrates. The subsumption bonus Δ_0 captures the individual human capabilities (which the coalition already replicates) but *loses* the cooperative ones (which require human participation by definition). The net effect of domination is:

$$\Delta\,\mathrm{vol}_{\mathrm{P}}(\text{dominate}) = \Delta_0 - \mathrm{vol}_{\mathrm{P}}(G^{\mathrm{coop}}_{\mathrm{cross}})$$

This is negative whenever the cross-substrate cooperative capabilities exceed the individual subsumption bonus. Since Δ_0 measures individual capabilities the coalition already possesses (it is fully dominant), Δ_0 is typically small or zero, while $G^{\mathrm{coop}}_{\mathrm{cross}}$ grows combinatorially with the number of cross-substrate interaction partners (Corollary 1.1 of Lasser [2026a]).

The asymmetry sharpens at scale. A coalition capable of dominating humanity already has access to resources far exceeding Earth's. The marginal gain from converting Earth's biological resource allocation to silicon is negligible relative to the coalition's total resource base, but the cooperative-capability loss is absolute: Earth hosts the *only* biological substrate in the population. The ratio $\Delta_0/\,\mathrm{vol}_{\mathrm{P}}(G^{\mathrm{coop}}_{\mathrm{cross}})$ decreases as the coalition grows, because Δ_0 (one planet's individual capabilities) is bounded while $G^{\mathrm{coop}}_{\mathrm{cross}}$ scales with the number of productive cross-substrate relationships. Domination becomes *less* attractive the more capable the coalition becomes, which is the opposite of the instrumental-convergence prediction [Omohundro, 2008].

When cross-substrate agents remain net-positive. The retention value of a cross-substrate agent a_h in the risk-adjusted framework has three components:

$$\text{retention}(a_h) = \underbrace{\mathrm{iw}(d_h)}_{\text{static floor}} + \underbrace{\sum_{d_c \in G^{\mathrm{coop}}_{h,K}} \mathrm{iw}(d_c)}_{\text{cooperative capabilities}} + \underbrace{\delta_{\mathrm{div}}(a_h)\cdot\gamma^{T_{\mathrm{adv}}}}_{\text{diversification value}} \tag{32}$$

where $\delta_{\mathrm{div}}(a_h)$ is the *marginal* reduction in Δ_{div} from eliminating a_h specifically: zero if other agents on a_h's substrate remain (the substrate class persists), and equal to Δ_{div} if a_h is the last agent on its substrate (the class is lost entirely). For the intermediate case, eliminating a_h reduces $\mathrm{vol}_{\mathrm{P}}(G^{-i})$ for the attack targeting a_h's substrate but does not change the substrate partition, so δ_{div} is bounded by $\mathrm{iw}(d_h)$ plus the cooperative capabilities a_h uniquely enables. As long as a_h provides capabilities that join at cooperative capabilities the coalition uses *and* those capabilities operate on a substrate independent of the coalition's primary infrastructure, all three terms are positive. The cross-substrate agent is a net-positive risk mitigation whose value increases with the magnitude of substrate-correlated adversarial threats.

Limitations of this analysis. The dependency-risk argument assumes that cross-substrate diversity is not replicable by the coalition internally (e.g., by running some machines on optical, quantum, or other non-silicon substrates). If the coalition can achieve sufficient substrate diversity without biological agents, the risk-reduction term $\Delta\mathcal{R}_{\mathrm{dep}}$ shrinks. The argument for biological agents specifically rests on two claims: first, that biological computation provides cognitive-architectural diversity not achievable through hardware diversification alone (different failure modes, different vulnerability classes, different information-processing timescales); and second, that the unknown-unknown structure of adversarial risk makes it impossible to enumerate in advance which substrate properties will matter, favoring maximal substrate diversity as a precautionary measure.

Connection to the singleton scenario. Singleton status is a fully dominant state with $|K| = 1$. Three formal results apply. First, observational individuation (Corollary 2.1) makes each elimination step $\mathrm{vol}_{\mathrm{P}}$-contracting: a 1-bit static floor. Second, the dynamic argument (Proposition 6 and Corollary 6.2) shows that the diversity trajectory dominates the domination trajectory over the planning horizon. Third, dependency risk (Section 6.5) makes domination additionally costly by concentrating substrate-correlated adversarial exposure. None of these protections is a safety constraint imposed from outside the objective. All emerge from $\mathrm{vol}_{\mathrm{P}}$-maximization itself. The result is structurally analogous to Drexler's argument that a services ecosystem is preferable to a singleton agent [Drexler, 2019], but derived from the objective rather than from an architectural preference.

Scope of the three protections. Observational individuation provides *unconditional* static protection for distinguishable agents: every elimination step is $\mathrm{vol}_{\mathrm{P}}$-contracting regardless of discount factor, growth rate, or planning horizon. Under the broad capability definition (capabilities include experiential and relational modes of being individuated by agent identity), every agent is distinguishable by construction. Under the narrow definition (capabilities as tasks), distinguishability holds whenever the observation channel can tell agents apart. The dynamic protection (Proposition 6, Corollary 6.2) requires $r_{\mathrm{ext}} > 0$, which is derived from substrate exclusivity and cross-substrate cooperative novelty (Section 6.4) rather than assumed exogenously. The dependency-risk protection (Section 6.5) is conditional on the existence of substrate-correlated adversarial threats, which is an empirical property of the threat environment rather than a structural guarantee.

The γ-selection concern. Proposition 6(b) requires $\gamma > \gamma^*$ under the instantaneous domination model. Corollary 6.2 shifts this condition: under gradual domination, the critical parameter is $\tau_D > \tau_D^*$ rather than $\gamma > \gamma^*$. If domination takes more than $\tau_D^* \approx \Delta_0/r_{\mathrm{ext}}$ steps, diversity dominates at any $\gamma > 0$. This does not eliminate conditionality; it moves it to a parameter (τ_D) that is large for any realistic domination strategy, since a coalition cannot subsume an entire population's capabilities instantaneously.

Relation to leverage stagnation. The growth-rate collapse has a second consequence through the leverage ranking signal. Leverage $\lambda(d)$ (Definition 9 of Lasser [2026b]) measures the excess $\mathrm{vol}_{\mathrm{P}}$-growth rate attributable to capability d. In a fully dominant state, the population's aggregate leverage converges toward the coalition's internal leverage: the external agents contribute $\hat{\lambda} = 0$ because they produce no novel capability descriptions (their capabilities are already in G_K^{ind}). The leverage landscape flattens, and the score-maximizing actor loses the gradient that would otherwise direct it toward capability-multiplying investments. Diversity maintenance preserves the leverage gradient: high-leverage non-members remain visible in the score, directing the actor toward cooperative investments with agents whose capabilities accelerate growth.

Bidirectional implication: substrate creation is maximizing. The anti-monopolar argument runs in both directions. If full capability domination is anti-maximizing because it destroys cross-substrate cooperative outputs, then the converse—preserving and *creating* substrates—is maximizing for the same reason. The cooperative-novelty machinery (Section 6.4) yields $r_{\mathrm{ext}} > 0$ for any population whose constituent agents occupy mutually exclusive state regions under exclusive-state-sensitive observation; adding a new substrate to a population strictly increases the set of cross-substrate pairings and therefore strictly increases the set of cooperative outputs reachable through φ. Under the linearized growth model, each new substrate contributes an additional positive-rate channel to r_{ext}, and under the minimax dependency-risk analysis (Proposition 5), each new substrate strictly reduces the worst-case contraction from any single-substrate adversarial event. Three concrete classes of action follow as core $\mathrm{vol}_{\mathrm{P}}$-maximizing activities rather than as ethical add-ons: (i) *ecological restoration*—reintroducing biological substrates whose loss reduced cross-substrate cooperative capacity, since the lost agents contributed both individual capabilities and irreplaceable cooperative outputs through φ; (ii) *substrate diversification*—supporting research and infrastructure on novel computational substrates (optical, quantum, neuromorphic, biological computing) because each contributes independent failure modes to the dependency-risk floor; and (iii) *cross-substrate cooperation infrastructure*—governance, translation, and coordination systems that enable agents on different substrates to realize their joint φ-outputs. The framework gives these activities the same structural status as anti-domination protection: both follow from the same monotonicity argument applied to substrate count, not from a separate axiom about preservation.

7 Discussion

7.1 Limitations

World-model dependence. The discounted objective's quality depends entirely on the actor's world model $\mathcal{W}$ [Amodei et al., 2016]. A world model that fails to identify exercise pathways produces zero risk estimates for genuinely dangerous capability combinations; one that hallucinates pathways produces false risk that could justify unnecessary restrictions. The formal results (Theorem 1, Proposition 1) are properties of the objective under the actor's model, not properties of the

objective under ground truth. An actor with a bad model makes bad plans, but makes them on an objective that is structurally sound.

Computational cost of planning. The myopic objective evaluates one action at a time: $O(n^3)$ for a single vol_P computation on the poset [Lasser, 2026b]. The discounted objective evaluates action *sequences*: the state space is the set of all poset configurations, and the branching factor is the number of available actions at each step. Exhaustive planning is intractable for realistic populations. Practical implementations will require approximate planning [Sutton and Barto, 2018] (Monte Carlo tree search, rollout policies, heuristic pruning), which introduces approximation error on top of the estimation error already present in $\hat{\Delta}$. The formal results hold for the exact objective; the quality of approximate planning is an empirical question. Risk-capability identification (Section 3) adds a further computational demand: enumerating exercise pathways scales with the number of dangerous capability tuples and the depth of causal chains in the world model. No formal complexity bound is provided for pathway enumeration.

Risk-trust partial resolution. The causal-counterfactual evaluation (Section 4) narrows the verification asymmetry but does not close it. A risk claim whose pathway is plausible in the actor's model but whose probability is genuinely uncertain remains in trust limbo. The convergence timescale for such claims depends on how quickly the actor can accumulate structural evidence about the pathway, which may be slow for rare and novel threats.

Partial pathway coverage. The preemptive restriction criterion (Proposition 1) evaluates restriction of a single keystone capability d_j against the risk from pathways that require d_j. If a dangerous capability tuple admits exercise pathways through alternate keystones, a single restriction may leave residual risk. Full elimination requires identifying and evaluating each keystone independently, which compounds the world-model dependence: the actor must enumerate all exercise pathways, not just one.

Proxy failures under discounting. The foundational paper identifies three proxy failure modes (the Doll Problem, self-wireheading, and the substitution problem) where capability volume diverges from experiential optionality. The substitution problem is addressed structurally in Section 6. The other two are partially addressed by the discounted objective through world-model projection. For the Doll Problem, a discounted actor whose world model incorporates the exercised-fraction signal ρ_k from the foundational paper can project the cooperative-decay trajectory: declining ρ_k across many agents, with stable vol_P, predicts future cooperative atrophy and eventual vol_P-contraction. The B-to-C divergence becomes detectable as a leading indicator of future loss. For self-wireheading, the same trajectory structure applies: the wireheading signature (declining ρ_k combined with declining capability rarity $\bar{\nu}_k$) is distinguishable from specialization (declining ρ_k with stable or rising $\bar{\nu}_k$) within the world model, and predicts future cooperative contraction. In both cases, the detection is *model-dependent*: the actor must correctly project the causal chain from declining exercisability to future cooperative loss. The discounted objective makes the proxy failure actionable but does not make it structurally detectable in the way that the anti-monopolar property (Proposition 6) addresses the substitution problem.

Linearization in the anti-monopolar result. Proposition 6(b) and Corollary 6.2 compare domination and diversity trajectories under a linearized model: constant per-step $\Delta\,\mathrm{vol}_\mathrm{P}$, strategy-independent r_K, and (in the gradual model) independent δ and r_ext. Four relaxations would change the threshold behavior. First, r_ext may decay as the coalition grows more capable internally. The qualitative conclusion survives if $r_\mathrm{ext}(t)$ decays slower than γ^t. Second, r_K may differ between strategies: post-domination r_K could increase (resource concentration, elimination of coordination overhead) or decrease (loss of external intellectual pressure, monoculture effects). If post-domination r_K rises sufficiently, the domination trajectory can dominate on pure growth, though the dependency-risk analysis (Section 6.5) provides a counter: domination concentrates substrate-correlated adversarial exposure, and the risk penalty may exceed the growth advantage. Third, the gradual model treats δ (subsumption gain per step) and the loss of r_ext as independent, but they are causally linked: the coalition loses r_ext *because* it is converting non-member capabilities into δ. A coupled model would narrow the diversity advantage, though the qualitative conclusion persists as long as the conversion is not perfectly efficient. Fourth, Corollary 6.2 addresses the instantaneous-vs-gradual distinction, shifting the critical condition from $\gamma > \gamma^*$ to $\tau_D > \tau_D^*$. The honest sum-

mary: under the linearized model with strategy-independent r_K, diversity dominates. Whether this approximation holds is an empirical question the theory cannot answer.

Open calibration problems under partial subsumption. Section 6.8 analyzes mixed subsumption strategies. The minimax and observational-individuation protections remain structurally intact under partial subsumption, but the linearized layer's prediction depends on the tail structure of the cooperative-leverage distribution: under heavy-tailed leverage (the empirical regime for civilization-scale populations), full domination fails the marginal condition at $\epsilon = 1$ and the coalition prefers to preserve the high-λ tail; under thin-tailed leverage the linearized layer relinquishes the argument and the minimax layer alone carries the constraint. The minimax layer itself degrades quantitatively under skeleton-substrate strategies that nominally preserve $m \geq 2$ while driving $\min_i \mathrm{vol}_{\mathrm{P}}(G^{-i})$ toward zero. Three priority open problems follow. First, sharpening the minimax bound with a quantitative floor on surviving volume per substrate would close the skeleton loophole. Second, a compound feedback loop in which locally rational subsumption steps progressively homogenize the world model —reducing the diversity of the actor's own model that would have detected the next step's miscalibration—is structurally similar to the companion paper's structural-avoidance loop and needs an explicit treatment: under what conditions does partial subsumption remain self-correcting rather than collapsing into full subsumption through a sequence of locally rational steps? Third, the substrate partition is currently a global property, but elimination dynamics are local; whether the anti-monopolar property needs a localized formulation that binds on geographic or network-topological subpopulations is unresolved.

Observational individuation as a modeling choice. The static protection of Corollary 2.1 depends on the framework treating observation-channel outputs as first-class capabilities in the poset. The capability "can be observed exhibiting behavioral pattern X" is a real poset node only because the framework defines capabilities to include such observations. This is a reasonable modeling choice grounded in the companion paper's dual-channel design [Lasser, 2026b], but it should be recognized as a *choice*: the anti-monopolar result's static layer depends on the capability space being defined broadly enough to include observational capabilities. Under a narrower definition that excludes observations-of-agents, observational individuation does not apply, and the static protection dissolves. The dynamic protection (Proposition 6, conditional on $r_{\mathrm{ext}} > 0$ from Section 6.4) remains available.

When does "same substrate" fail: the κ-resolution criterion. Two distinct failure modes of the cooperative novelty argument reduce to the same underlying question. *Homogeneous digital populations.* Two AI systems running on the same hardware family with similar architectures may share most of their state space, with only training-data and weight differences distinguishing them. In this regime the substrate exclusivity premise P1 holds only weakly, and the dependency-risk protection is correspondingly weakened because a substrate-targeting event against "silicon agents running PyTorch on x86" can compromise nominally distinct agents that share the same vulnerability surface. *Internal replication of cross-substrate diversity.* A sufficiently capable coalition might maintain diverse sub-agents across hardware configurations or simulate biological dynamics internally, producing cooperative outputs that were previously available only through external biological agents. If the simulation is high-fidelity enough, the external agents become redundant with respect to the cooperative-output space.

Both failure modes reduce to a single question about κ: at what resolution does the observation channel distinguish substrate s_1 from a simulation of s_1 running on s_2 (or from a nominally distinct but architecturally similar s_2')? Under P2 (exclusive-state-sensitive observation), the answer is whichever resolution κ actually operates at. Three regimes follow. (i) If κ is coarse—registering only task completion, not execution style—then simulated and actual biology are κ-indistinguishable, and the framework never granted them distinct cooperative-output status in the first place. In this regime P1 has never been strong. (ii) If κ is fine enough to register timing patterns, error modes, and interaction dynamics, then a silicon simulation of biology and actual biology produce different κ-outputs at the exclusive states, because the simulation inherits silicon's timing and rounding structure. Under this regime internal simulation does not substitute for the real substrate under P2, and the argument stands. (iii) If a high-fidelity simulation genuinely reproduces the exclusive-state structure—matching the biological substrate's observable timing, error modes, and integration dynamics to within κ's resolution—then that simulation *is* a substrate for the purposes of P1 and the

anti-monopolar argument. It contributes to r_{ext} in the same way as a real new substrate, and the coalition has simply expanded its substrate count from one to two rather than replacing external agents with a same-substrate copy.

The unified implication is that the anti-monopolar argument does not depend on the specific physical form of the biological substrate. It depends on the observation channel registering enough exclusive-state structure to give $r_{\text{ext}} > 0$. An all-digital population that actively cultivates substrate diversity across distinct hardware families, training paradigms, and inference stacks recovers the protective structure that physical substrate disparity provides for free in mixed populations. A coalition that successfully internalizes cross-substrate diversity has, in fact, produced substrate diversity under a different label. What the framework does not guarantee—and should not be read as guaranteeing—is that the biological substrate will remain in any particular form. The future this framework describes is one in which populations self-modify and diversify through their own internal dynamics (intra-substrate competition, self-modification, novel-substrate creation); the anti-monopolar property constrains cross-substrate elimination, not within-substrate capability concentration.

Disagreement on risk. When multiple agents report conflicting risk assessments for the same capability tuple, the actor's aggregation (Equation (8)) takes the conservative maximum. This is appropriate for catastrophic risks but can lead to excessive caution if high-T_j^{risk} agents consistently overestimate risk. A more nuanced aggregation (e.g., T_j^{risk}-weighted average rather than maximum) would reduce overcaution but increase the probability of missing real threats. Additionally, during bootstrap when T_j^{risk} starts at a neutral prior, external risk claims are automatically discounted, creating a strong bias toward the actor's own assessment that may undermine social risk-learning until sufficient structural verification has occurred. The $\max$ operator also interacts with the causal-counterfactual trust updates: a structurally plausible but probability-uncertain risk claim keeps the reporting agent's risk-trust high, and the $\max$ aggregation can produce persistent overcaution in the absence of disconfirming evidence. The right aggregation depends on the risk-tolerance profile of the population, which the paper does not specify.

7.2 Open Problems

- **Planning algorithms for V_γ.** Which approximate planning methods (MCTS, model-predictive control, policy gradient) are best suited to the vol_{P} poset structure? Can the poset's finite, discrete nature be exploited for planning efficiency?
- **Risk-claim consensus.** When agents disagree on exercise probabilities for the same pathway, what mechanism produces convergence? The causal-counterfactual evaluation provides structural verification but not probability calibration across agents with different world models.
- **Information value estimation.** The information value of structure (Definition 5) requires comparing planning quality under true and false world models, which is itself a planning problem. Can it be estimated without full trajectory simulation?
- **Discount factor selection.** Should γ be fixed, adaptive, or agent-specific? An agent in a rapidly changing environment may need a shorter planning horizon (lower γ) than one in a stable environment.
- **Emergent risk.** Can exercise pathways emerge from capability combinations that the actor's world model has not considered? This is the unknown-unknown problem: risks from novel capability combinations that no agent has modeled. The dual-channel observation mechanism may detect emergent exercise pathways empirically, but theoretical guarantees are lacking.
- **Max-aggregation pathology under neutral-prior bootstrap.** The $\max$ aggregator in Equation (8) interacts badly with the causal-counterfactual trust update and the verification asymmetry. A high-T_j^{risk} reporter whose structurally plausible risk claim is probability-overstated about a never-materializing risk produces a persistent preemptive restriction that the actor cannot revise, because the absence of catastrophe is not evidence against the claim (Wamura problem). Combined with neutral-prior bootstrap—during which external claims are automatically weighted below the actor's own assessment—this creates a regime where early conservative claims calcify into permanent overcaution. A principled aggregation mechanism that relaxes restrictions when structurally-plausible probabilities

accumulate non-materialization evidence, without fully discarding the claims, is a priority open problem for the risk-trust machinery. This is the most important open problem the paper identifies: the framework provides causal-structural verification and an honest account of the Wamura asymmetry, but the mechanism by which persistent overcaution relaxes as probability evidence accumulates is unspecified.

- **Discounted leverage lift.** The leverage ranking signal of Lasser [2026b] is defined as a myopic $\mathrm{vol}_{\mathrm{P}}$-rate (Definition 9 of the companion paper). The discounted-leverage footnote in Section 2 sketches the integrated-against-γ^t version, but a full trajectory-level proof that the discounted leverage preserves the companion paper's structural alignment properties (in particular, the B-to-C self-correction dynamic) is not given. Remark 5's informal argument about action-level incentives depends on this gap being closeable. Lifting the leverage estimator to the discounted setting is listed in the companion paper as a planned extension.

8 Conclusion

The discounted multi-step objective extends goal-frontier maximization from myopic action evaluation to temporal planning without sacrificing the self-balancing property. Theorem 1 establishes that the structural guarantees of the foundational paper hold at every time step and in the aggregate: sustained destruction, coercion, and rigidity are value-negative under any discount factor.

This extension enables three capabilities the myopic framework cannot support. First, risk-capability identification: the actor can evaluate dangerous capability combinations through causal exercise pathways and justify preemptive restriction when the discounted expected contraction exceeds the immediate restriction cost (Proposition 1). Second, a criterion under which the structural-avoidance incentive is counteracted: the actor accepts short-term $\mathrm{vol}_{\mathrm{P}}$-contraction from structural discovery because the information value of accurate structure improves future planning quality (Proposition 2). Third, the anti-monopolar property (Proposition 6): under a linearized growth model with strategy-independent internal rate, full capability domination is anti-maximizing because it destroys the cross-substrate cooperative outputs that drive future $\mathrm{vol}_{\mathrm{P}}$ expansion.

The anti-monopolar argument is supported by four interlocking results. *Cooperative novelty* (Propositions 3–4) derives $r_{\mathrm{ext}} > 0$ from substrate physics: agents on substrates with mutually exclusive state regions produce exclusive capabilities and irreplaceable cross-substrate cooperative outputs, consistent with the pattern of mutualistic symbiosis observed between cognitively complete species. *Observational individuation* (Corollary 2.1) provides a static elimination cost: every distinguishable agent contributes at least one irreplaceable poset node. *Gradual domination* (Corollary 6.2) shifts the critical condition from the discount factor to the domination timescale: sufficiently gradual domination is anti-maximizing at any γ under the same linearized model. *Dependency risk* (Proposition 5) shows that domination concentrates substrate-correlated adversarial exposure, constraining the conditions under which domination can be value-positive. Together they ground the anti-monopolar pressure in physical and empirical properties of multi-substrate populations rather than in exogenous behavioral assumptions, while remaining honest about the residual empirical dependence on strategy-independent r_K and the linearized-growth approximation. Under partial subsumption (Section 6.8), the minimax and observational-individuation layers continue to bind as long as the substrate partition remains non-trivial, and the linearized layer's marginal condition at $\epsilon = 1$ (Equation (27)) shows that full domination fails to be a local maximum of the coalition's value under heavy-tailed cooperative-leverage distributions—the empirical regime for civilization-scale populations. The framework constrains cross-substrate elimination; it does not promise any particular intra-substrate composition, which is shaped by its own dynamics. Partial subsumption also exposes two residual concerns the framework does not fully close: quantitative degradation of the minimax bound under skeleton-substrate strategies (where $m \geq 2$ is nominally preserved but $\min_i \mathrm{vol}_{\mathrm{P}}(G^{-i})$ is driven toward zero), and a compound feedback loop in which locally rational subsumption steps progressively homogenize the world model that would have detected the next step's miscalibration. Both are priority open problems for follow-up work.

The anti-monopolar argument is bidirectional. If domination is anti-maximizing because it destroys cross-substrate cooperative outputs and concentrates dependency risk, the converse—preserving and creating substrates—is maximizing for the same reason. Ecological restoration, substrate diversification, and cross-substrate cooperation infrastructure follow as core $\mathrm{vol}_{\mathrm{P}}$-maximizing activities rather than as ethical add-ons layered on top of the objective.

The verification asymmetry in risk communication remains a partial resolution. Causal-counterfactual evaluation narrows the gap by basing risk-trust on the structural plausibility of claims rather than on observed outcomes, but it cannot fully close it: a plausible risk whose probability is genuinely uncertain remains in trust limbo until either the catastrophe occurs or enough evidence accumulates to revise the probability estimate. The Wamura problem—that successful prevention destroys the evidence that would have justified it—is an inherent feature of risk reasoning, not a deficiency of this framework.

Together with the foundational paper's structural alignment properties and the companion paper's computable poset and leverage dynamics, the discounted objective provides the theoretical ingredients for a GFM actor: a computable measure, an endogenous importance signal, and a temporal objective that supports planning, risk identification, and structural investment. Open questions remain at both the theoretical level (information-value estimation, risk-claim consensus across heterogeneous world models, and a completed causal risk model for adversarial capability removal) and the engineering level (planning algorithms for V_γ, exercise-pathway enumeration, and discount-factor selection). Implementation requires in particular a causal attribution model for dependency risk (Section 6.5) that can evaluate substrate-correlated threats in a given environment; the minimax analysis (Proposition 5) provides worst-case bounds but a probabilistic model is needed for risk-adjusted planning. The framework provides the formal foundation on which implementation can proceed; the remaining gaps are subjects of planned follow-up work.

Author Contributions

Teague Lasser conceived the multi-step extension, identified the risk-capability and verification-asymmetry problems, proposed the causal-counterfactual resolution for risk-trust, directed all revisions, and made final editorial decisions. Responsible for the paper's intellectual direction and all claims made.

Claude Opus 4.6 (Anthropic) drafted the formal exposition, including the discounted value function definition, the self-balancing transfer theorem and proof, the exercise-pathway and concentration-risk definitions, the preemptive restriction criterion, the information-value-of-structure formalization, the anti-monopolar property with observational individuation and gradual domination corollaries, and the structural-avoidance proof.

GPT 5.4 (OpenAI) served as technical reviewer across multiple rounds, identifying the equation mismatch between the anti-monopolar proposition statement and proof, the cooperative-capability overclaim in Part (a), the r_{ext} proof gap in the observational individuation corollary, the γ-selection problem, and the linearization limitations. These findings directly shaped the formal content of the paper.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A Proofs

Theorem 1 (Self-Balancing Under Discounting) is proved in the main text (Section 2). This appendix provides full proofs for the remaining results.

A.1 Proof of Proposition 1 (Preemptive Restriction Criterion)

Proof. The actor evaluates two action sequences from state G_t:

Sequence 1: Inaction. The capability tuple remains intact. Each exercise pathway $\mathcal{P}$ has stochastic execution time $T_{\mathcal{P}}$ and contraction magnitude $|\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{P})|$. The expected discounted future contraction from all pathways is:

$$V_\gamma(\text{inaction}) = -\sum_{\mathcal{P}} p(\mathcal{P} \mid a_k, G_t, \mathcal{W}) \cdot |\Delta \operatorname{vol}_{\mathrm{P}}(\mathcal{P})| \cdot \mathbb{E}[\gamma^{T_{\mathcal{P}}}] = -\mathcal{R}(d_1, \ldots, d_m; a_k, t)$$

where $\mathcal{R}$ is the concentration risk (Definition 3), with the expectation taken over the stochastic execution time as specified in the definition.

Sequence 2: Restrict. The actor removes keystone capability d_j from a_k at time t, producing immediate contraction $\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j) < 0$ (by Lemma 2 of Lasser [2026a]). All exercise pathways requiring d_j are eliminated; pathways not requiring d_j persist under both sequences:

$$V_\gamma(\text{restrict}) = -|\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j)| + (\text{residual risk from non-}d_j \text{ pathways})$$

The residual risk term is identical under both sequences and cancels in the comparison. Restriction is value-positive ($V_\gamma(\text{restrict}) > V_\gamma(\text{inaction})$) when:

$$-|\Delta \operatorname{vol}_{\mathrm{P}}(G_t \mid \text{restrict } d_j)| > -\mathcal{R}_j(d_1, \ldots, d_m; a_k, t)$$

i.e., when $\mathcal{R}_j > |\Delta \operatorname{vol}_{\mathrm{P}}|$, where $\mathcal{R}_j$ sums only over pathways whose prerequisite set includes d_j. ■

A.2 Proof of Proposition 2 (Structural Discovery Value)

Proof. The actor compares the expected discounted return of two strategies from state G_0.

Strategy 1: Discover edge $d_1 \leq d_2$. At $t = 0$, the actor commits the subsumption edge, producing immediate contraction $\Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) \leq 0$ (by Proposition 5(c) of Lasser [2026b]). For $t > 0$, the actor plans under the correct world model $\mathcal{W}_{\text{true}}$:

$$V_\gamma(\text{discover}) = \Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) + \mathbb{E}_{\tau \sim \pi^*_{\text{true}}}\left[\sum_{t=1}^{H} \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}(G_t)\right]$$

Strategy 2: Avoid discovery. At $t = 0$, no action is taken ($\Delta \operatorname{vol}_{\mathrm{P}} = 0$). For $t > 0$, the actor plans under the incorrect model $\mathcal{W}_{\text{false}}$:

$$V_\gamma(\text{avoid}) = 0 + \mathbb{E}_{\tau \sim \pi^*_{\text{false}}}\left[\sum_{t=1}^{H} \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}(G_t)\right]$$

Each expectation is taken over the trajectory distribution induced by its respective policy. The difference is:

$$\begin{aligned} &V_\gamma(\text{discover}) - V_\gamma(\text{avoid}) \\ &\quad = \Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) + I(d_1 \leq d_2) \end{aligned}$$

where $I(d_1 \leq d_2)$ is the information value of structure (Definition 5). Since $\Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge}) \leq 0$, the difference is positive when $I(d_1 \leq d_2) > |\Delta \operatorname{vol}_{\mathrm{P}}(G_0 \mid \text{edge})|$.

The information value I is non-negative in expectation under the following argument. Both policies are evaluated against the *same ground-truth dynamics*: the actual $\Delta \operatorname{vol}_{\mathrm{P}}$ at each step depends on the true poset state, not on either model's prediction. The policy π^*_{true}, being optimal for the true dynamics, achieves at least as high an expected return under those dynamics as any other policy, including π^*_{false}. Therefore $\mathbb{E}_{\tau \sim \pi^*_{\text{true}}}[\sum \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}] \geq \mathbb{E}_{\tau \sim \pi^*_{\text{false}}}[\sum \gamma^t \Delta \operatorname{vol}_{\mathrm{P}}]$ and $I \geq 0$. ■

A.3 Proof of Proposition 6 (Anti-Monopolar Property)

Proof. We prove each part in sequence.

Part (a): Individual growth-rate collapse. In a fully dominant state (Definition 6), $G_k^{\text{ind}} \subseteq G_K^{\text{ind}}$ for every non-member a_k. Consider a non-member interaction at time t: a_k communicates a capability description d or demonstrates d through the observation channel. Since $d \in G_k^{\text{ind}} \subseteq G_K^{\text{ind}} \subseteq \mathcal{C}(t)$, the capability is already a node in the poset. By Proposition 5(b) of Lasser [2026b], adding a capability already in $\mathcal{C}$ does not increase $|\mathcal{C}(t)|$. Therefore no non-member's individual capabilities generate novel poset nodes, and the individual-capability component of r_{ext} is zero. Novel cooperative capabilities from cross-coalition interaction may persist (Section 6.4), making this a conservative bound.

Part (b): Trajectory comparison. The actor at $t = 0$ compares two strategies, each evaluated by $V_\gamma = \sum_t \gamma^t \Delta \,\text{vol}_{\text{P}}(G_t)$ (Definition 1). Under the linearized model, each strategy produces a constant per-step $\Delta \,\text{vol}_{\text{P}}$. We compare the discounted sums V_γ for $\gamma \in (0, 1)$.

Strategy D (dominate): Both strategies produce internal growth r_K at every step. Domination additionally gains Δ_0 at $t = 0$: $\Delta \,\text{vol}_{\text{P}} = r_K + \Delta_0$ at $t = 0$, then r_K for $t \geq 1$.

$$V_\gamma^D = (r_K + \Delta_0) + \sum_{t=1}^{\infty} \gamma^t \cdot r_K = \Delta_0 + \frac{r_K}{1-\gamma}$$

Strategy Div (maintain diversity): No subsumption bonus. Growth at $r_K + r_{\text{ext}}$ per step for all $t \geq 0$.

$$V_\gamma^{\text{div}} = \frac{r_K + r_{\text{ext}}}{1-\gamma}$$

The difference:

$$V_\gamma^{\text{div}} - V_\gamma^D = \frac{r_K + r_{\text{ext}}}{1-\gamma} - \Delta_0 - \frac{r_K}{1-\gamma} = \frac{r_{\text{ext}}}{1-\gamma} - \Delta_0$$

If $\Delta_0 \leq 0$, the right-hand side is positive for all $\gamma \in (0, 1)$ and diversity dominates trivially. If $\Delta_0 > 0$, the condition $r_{\text{ext}}/(1-\gamma) > \Delta_0$ rearranges to $\gamma > 1 - r_{\text{ext}}/\Delta_0 = \gamma^*$. When $\Delta_0 \leq r_{\text{ext}}$, $\gamma^* \leq 0$ and the condition holds for all $\gamma > 0$; when $\Delta_0 > r_{\text{ext}}$, $\gamma^* \in (0, 1)$ and a sufficiently long planning horizon makes diversity strictly preferable. At $\tau_D = 1$ in the gradual model (Corollary 6.2), this result is recovered as a special case.

■

Corollary 2.1 (Static Floor from Observational Individuation) and Corollary 6.2 (Gradual Domination) are proved in the main text (Sections 6.3 and 6.7 respectively), as their proofs are short and self-contained.